

$\int_a^b f(x)dx$
 $\lim_{x \rightarrow 0} f(x)$
 $\frac{dy}{dx}$
 $ax + by \leq c$
 $\sqrt{x^2 + y^2}$

UNIT 1

FUNCTIONS AND LIMITS

Function:

A **Function** f from a set X to a set Y is a rule or correspondence that assigns to each element x in X a unique element y in Y .

Symbolically we write it as $f : X \rightarrow Y$ and read as f is a function from X to Y .

Domain and Range of Function:

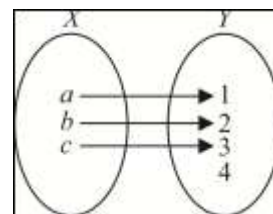
If f is a function from X to Y then

X is called the domain of f and the set of corresponding elements in Y is called range of f

For example:

Domain = $\{a, b, c\}$

Range = $\{1, 2, 3\}$



Example 1: Given $f(x) = x^3 - 2x^2 + 4x - 1$, find

- (i) $f(0)$ (ii) $f(1)$ (iii) $f(-2)$ (iv) $f(1+x)$ (v) $f\left(\frac{1}{x}\right), x \neq 0$

Solution:

$$f(x) = x^3 - 2x^2 + 4x - 1$$

(i) $f(0) = (0)^3 - 2(0)^2 + 4(0) - 1 = 0 - 0 + 0 - 1 = -1$

(ii) $f(1) = (1)^3 - 2(1)^2 + 4(1) - 1 = 1 - 2 + 4 - 1 = 2$

(iii) $f(-2) = (-2)^3 - 2(-2)^2 + 4(-2) - 1 = -8 - 8 - 8 - 1 = -25$

(iv) $f(1+x) = (1+x)^3 - 2(1+x)^2 + 4(1+x) - 1$
 $= 1 + 3x + 3x^2 + x^3 - 2 - 4x - 2x^2 + 4 + 4x - 1$
 $= x^3 + x^2 + 3x + 2$

(v) $f\left(\frac{1}{x}\right) = \left(\frac{1}{x}\right)^3 - 2\left(\frac{1}{x}\right)^2 + 4\left(\frac{1}{x}\right) - 1 = \frac{1}{x^3} - \frac{2}{x^2} + \frac{4}{x} - 1, x \neq 0$

Example 2: Let $f(x) = x^2$. Find the domain and range of f .

Solution:

$f(x)$ is defined for every real number x .

Further for every real number x , $f(x) = x^2$ is a non-negative real number. So

Domain f = Set of all real numbers.

Range f = Set of all non-negative real numbers.

Example 3: Let $f(x) = \frac{x}{x^2 - 4}$. Find the domain and range of f .

Solution:

$$f(x) = \frac{x}{x^2 - 4}$$

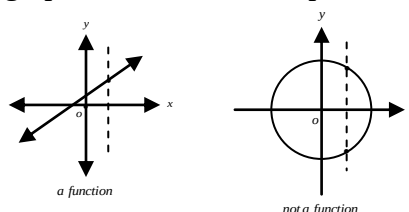
$f(x)$ is not defined if $x^2 - 4 = 0 \Rightarrow x^2 = 4$ or $x = \pm 2$

Domain f = Set of all real numbers except -2 and 2 .

Range f = Set of all real numbers.

Vertical Line Test:

If a vertical line meets a graph in more than one point, then it is not a graph of a function.



Piece –Wise (Compound) Function:

A function which is defined by two or more than two rules is called Piece-wise function.

For example:

$$f(x) = \begin{cases} x & \text{if } 0 \leq x \leq 1 \\ x-1 & \text{if } 1 < x \leq 2 \end{cases}$$

Algebraic Functions:

Algebraic functions are those functions which are defined by algebraic expressions.

For example:

$$f(x) = 3x + 5, f(x) = x^2 + 3x + 2$$

We classify Algebraic functions as follows:

(i) Polynomial Function:

A function P of the form $P(x) = a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \dots + a_2 x^2 + a_1 x + a_0$

for all x , where the coefficients $a_n, a_{n-1}, \dots, a_2, a_1, a_0$ are real numbers and the exponents are non-negative integers, is called a **polynomial function**. If $a_n \neq 0$ then $P(x)$ is called a **polynomial function** of degree n and a_n is the leading co-efficient of $P(x)$.

For example:

$P(x) = 2x^4 - 3x^3 + 2x - 1$ is a **polynomial function** of degree 4 with leading coefficient 2.

(ii) Linear Function:

If the degree of a polynomial function is 1, then it is called a **linear function**.

Symbolically we write $f(x) = ax + b$ where $a \neq 0$, a, b are real numbers.

For example:

$f(x) = 3x + 4$, $f(x) = x + 2$ are **linear functions** of x .

(iii) Identity Function:

For any set X , a function $I : X \rightarrow X$ of the form $I(x) = x \quad \forall x \in X$ is called an **identity function**.

(iv) Constant Function:

Let X and Y be sets of real numbers. A function $C : X \rightarrow Y$ defined by $C(x) = a, \forall x \in X, a \in Y$ and fixed is called **constant function**. For example $C : R \rightarrow R$ defined by $C(x) = 2, \forall x \in R$ is a constant function.

(v) Rational Function:

A function $R(x)$ of the form $\frac{P(x)}{Q(x)}$, where both $P(x)$ and $Q(x)$ are polynomial functions and $Q(x) \neq 0$, is called a **rational function**.

Exponential Function:

A function, in which the variable appears as exponent (power), is called an **exponential function**. The functions $y = e^{ax}, y = e^x, y = 2^x = e^{x \ln 2}$, etc are exponential functions of x .

Logarithmic Function:

If $x = a^y$, then $y = \log_a x$, where $a > 0, a \neq 1$ is called **Logarithmic function** of x .

- (i) If $a = 10$, then we have $\log_{10} x$ (written as $\log x$) which is known as the **common logarithm** of x .
- (ii) If $a = e$, then we have $\log_e x$ (written as $\ln x$) which is known as the **natural logarithm** of x .

Hyperbolic Functions:

- (i) $\sinh x = \frac{1}{2}(e^x - e^{-x})$ is called **hyperbolic sine** function. Its domain and range are the set of all real numbers.
- (ii) $\cosh x = \frac{1}{2}(e^x + e^{-x})$ is called **hyperbolic cosine** function. Its domain is the set of all real numbers and the range is the set of all numbers in the interval $[1, +\infty)$.
- (iii) The remaining four hyperbolic functions are defined in terms of the hyperbolic sine and the hyperbolic cosine function as follows:

$$\begin{aligned} \tanh x &= \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}} & ; & \quad \operatorname{sech} x = \frac{1}{\cosh x} = \frac{2}{e^x + e^{-x}} \\ \coth x &= \frac{\cosh x}{\sinh x} = \frac{e^x + e^{-x}}{e^x - e^{-x}} & ; & \quad \operatorname{csch} x = \frac{1}{\sinh x} = \frac{2}{e^x - e^{-x}} \end{aligned}$$

Inverse Hyperbolic Functions:

The inverse hyperbolic functions are expressed in terms of natural logarithms and we shall study them in higher classes.

$$(i) \quad \sinh^{-1} x = \ln(x + \sqrt{x^2 + 1}), \text{ for all } x \quad (ii) \quad \cosh^{-1} x = \ln(x + \sqrt{x^2 - 1}), x \geq 1$$

$$(iii) \quad \tanh^{-1} x = \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right), |x| < 1 \quad (iv) \quad \coth^{-1} x = \frac{1}{2} \ln\left(\frac{x+1}{x-1}\right), |x| < 1$$

$$(v) \quad \operatorname{sech}^{-1} x = \ln\left(\frac{1}{x} + \frac{\sqrt{1-x^2}}{x}\right), 0 < x \leq 1 \quad (vi) \quad \operatorname{cosech}^{-1} x = \ln\left(\frac{1}{x} + \frac{\sqrt{1+x^2}}{|x|}\right), x \neq 0$$

Explicit Function:

If y is easily expressed in terms of the independent variable x , then y is called an **explicit function** of x .

For example:

$$y = x^2 + 2x - 1, y = \sqrt{x-1} \text{ are explicit functions of } x.$$

Symbolically it can be written as $y = f(x)$.

Implicit Function:

If x and y are so mixed up and y cannot be expressed in terms of the independent variable x , then y is called an **implicit function** of x . For example,

$$x^2 + xy + y^2 = 2, \frac{xy^2 - y + 9}{xy} = 1 \text{ are implicit functions of } x \text{ and } y.$$

Symbolically it is written as $f(x, y) = 0$.

Parametric Functions:

Sometimes a curve is described by expressing both x and y as function of a third variable " t " or " θ " which is called a parameter. The equations of the type $x = f(t)$ and $y = g(t)$ are called the parametric equations of the curve.

The functions of the form:

$$(i) \quad \begin{matrix} x = at^2 \\ y = at \end{matrix} \quad (ii) \quad \begin{matrix} x = a \cos t \\ y = a \sin t \end{matrix} \quad (iii) \quad \begin{matrix} x = a \cos \theta \\ y = b \sin \theta \end{matrix} \quad (iv) \quad \begin{matrix} x = a \sec \theta \\ y = a \tan \theta \end{matrix}$$

are called **parametric functions**. Here the variable t or θ is called parameter.

Even Function:

A function f is said to be an **even function** if $f(-x) = f(x)$, for every number x in the domain of f .

For example:

$$f(x) = x^2, f(x) = \cos x \text{ are even functions of } x.$$

Odd Function:

A function f is said to be an **odd function** if $f(-x) = -f(x)$, for every number x in the domain of f .

For example:

$$f(x) = \sin x, f(x) = x^3 \text{ are odd functions of } x.$$

Some Important Results**Hyperbolic Identities:**

- $\cosh^2 x + \sinh^2 x = \cosh 2x$
- $\cosh^2 x - \sinh^2 x = 1$
- $2 \sinh x \cosh x = \sinh 2x$
- $1 - \tanh^2 x = \operatorname{sech}^2 x$
- $\coth^2 x - 1 = \operatorname{cosech}^2 x$

Parametric Equations:

- $x = a \cos \theta$
 $y = a \sin \theta$ represent the equation of circle $x^2 + y^2 = a^2$
- $x = a \cos \theta$
 $y = b \sin \theta$ represent the equation of ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.
- $x = a \sec \theta$
 $y = b \tan \theta$ represent the equation of hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.
- $x = at^2$
 $y = 2at$ represent the equation of parabola $y^2 = 4ax$.

EXERCISE 1.1**Q.1 Given that:**

(a) $f(x) = x^2 - x$

(b) $f(x) = \sqrt{x+4}$

Find

(i) $f(-2)$

(ii) $f(0)$

(iii) $f(x-1)$

(iv) $f(x^2 + 4)$

(a) $f(x) = x^2 - x$

(i) $f(-2)$

Solution:

$$f(x) = x^2 - x$$

Put $x = -2$

$$f(-2) = (-2)^2 - (-2)$$

$$= 4 + 2$$

$$\boxed{f(-2) = 6}$$

(ii) $f(0)$

Solution:

$$f(x) = x^2 - x$$

Put $x = 0$

$$f(0) = 0^2 - 0$$

$$\boxed{f(0) = 0}$$

(iii) $f(x-1)$

Solution:

$$f(x) = x^2 - x$$

Replace x by $x-1$

$$f(x-1) = (x-1)^2 - (x-1)$$

$$= x^2 - 2x + 1 - x + 1$$

$$\boxed{f(x-1) = x^2 - 3x + 2}$$

(iv) $f(x^2 + 4)$

Solution:

$$f(x) = x^2 - x$$

Replace x by $x^2 + 4$

$$f(x^2 + 4) = (x^2 + 4)^2 - (x^2 + 4)$$

$$= x^4 + 8x^2 + 16 - x^2 - 4$$

$$f(x^2 + 4) = x^4 + 7x^2 + 12$$

(b) $f(x) = \sqrt{x+4}$

(i) $f(-2)$

Solution:

$$f(x) = \sqrt{x+4}$$

Put $x = -2$

$$f(-2) = \sqrt{-2+4}$$

$$f(-2) = \sqrt{2}$$

(ii) $f(0)$

Solution:

$$f(x) = \sqrt{x+4}$$

Put $x = 0$

$$f(0) = \sqrt{0+4} = \sqrt{4}$$

$$f(0) = 2$$

(iii) $f(x-1)$

Solution:

$$f(x) = \sqrt{x+4}$$

Replace x by $x-1$

$$f(x-1) = \sqrt{x-1+4}$$

$$f(x-1) = \sqrt{x+3}$$

(iv) $f(x^2 + 4)$

Solution:

$$f(x) = \sqrt{x+4}$$

Replace x by $x^2 + 4$

$$f(x^2 + 4) = \sqrt{x^2 + 4 + 4}$$

$$f(x^2 + 4) = \sqrt{x^2 + 8}$$

Q.2 Find $\frac{f(a+h) - f(a)}{h}$ and simplify

where,

(i) $f(x) = 6x - 9$

(ii) $f(x) = \sin x$

(iii) $f(x) = x^3 + 2x^2 - 1$

(iv) $f(x) = \cos x$

(i) $f(x) = 6x - 9$

Solution:

$$f(x) = 6x - 9$$

$$\frac{f(a+h) - f(a)}{h}$$

$$= \frac{[6(a+h) - 9] - (6a - 9)}{h}$$

$$= \frac{6a + 6h - 9 - 6a + 9}{h}$$

$$= \frac{6h}{h}$$

$$\frac{f(a+h) - f(a)}{h} = 6$$

(ii) $f(x) = \sin x$

Solution:

$$f(x) = \sin x$$

$$\frac{f(a+h) - f(a)}{h}$$

$$= \frac{\sin(a+h) - \sin a}{h}$$

$$= \frac{1}{h} [\sin(a+h) - \sin a]$$

$$\because \sin P - \sin Q = 2 \cos \left(\frac{P+Q}{2} \right) \sin \left(\frac{P-Q}{2} \right)$$

$$= \frac{1}{h} \left[2 \cos \left(\frac{a+h+a}{2} \right) \sin \left(\frac{a+h-a}{2} \right) \right]$$

$$= \frac{2}{h} \left[\cos \left(\frac{2a+h}{2} \right) \sin \left(\frac{h}{2} \right) \right]$$

$$\frac{f(a+h) - f(a)}{h} = \frac{2}{h} \left[\cos \left(a + \frac{h}{2} \right) \sin \left(\frac{h}{2} \right) \right]$$

(iii) $f(x) = x^3 + 2x^2 - 1$

Solution:

$$f(x) = x^3 + 2x^2 - 1$$

$$\frac{f(a+h) - f(a)}{h}$$

$$\begin{aligned}
&= \frac{[(a+h)^3 + 2(a+h)^2 - 1] - [a^3 + 2a^2 - 1]}{h} \\
&= \frac{a^3 + 3a^2h + 3ah^2 + h^3 + 2(a^2 + 2ah + h^2) - 1 - a^3 - 2a^2 + 1}{h} \\
&= \frac{3a^2h + 3ah^2 + h^3 + 2a^2 + 4ah + 2h^2 - 2a^2}{h} \\
&= \frac{3a^2h + 3ah^2 + h^3 + 4ah + 2h^2}{h} \\
&= \frac{h[3a^2 + 3ah + h^2 + 4a + 2h]}{h} \\
&\quad \boxed{\frac{f(a+h) - f(a)}{h} = 3a^2 + 3ah + h^2 + 4a + 2h}
\end{aligned}$$

(iv) $f(x) = \cos x$

Solution:

$$\begin{aligned}
&f(x) = \cos x \\
&\frac{f(a+h) - f(a)}{h} \\
&= \frac{\cos(a+h) - \cos a}{h} \\
&= \frac{1}{h} [\cos(a+h) - \cos a] \\
&\because \cos P - \cos Q = -2 \sin\left(\frac{P+Q}{2}\right) \sin\left(\frac{P-Q}{2}\right) \\
&= \frac{1}{h} \left[-2 \sin\left(\frac{a+h+a}{2}\right) \sin\left(\frac{a+h-a}{2}\right) \right] \\
&= -\frac{2}{h} \left[\sin\left(\frac{2a+h}{2}\right) \sin\left(\frac{h}{2}\right) \right] \\
&\quad \boxed{\frac{f(a+h) - f(a)}{h} = -\frac{2}{h} \left[\sin\left(a + \frac{h}{2}\right) \sin\left(\frac{h}{2}\right) \right]}
\end{aligned}$$

Q.3 Express the following:

(a) The perimeter P of square as a function of its area A .

Solution:

Let x be the length of each side of square, then

$$A = x^2 \Rightarrow \sqrt{A} = \sqrt{x^2}$$

$$x = \sqrt{A}$$

$$P = 4x \dots (i)$$

Put $x = \sqrt{A}$ in equation (i)

$$\boxed{P = 4\sqrt{A}}$$

(b) The area A of a circle as a function of its circumference C .

Solution:

Let r be the radius of circle, then

$$A = \pi r^2 \dots (i)$$

$$C = 2\pi r \Rightarrow r = \frac{C}{2\pi}$$

Put $r = \frac{C}{2\pi}$ in equation (i)

$$A = \pi \left(\frac{C}{2\pi} \right)^2$$

$$A = \pi \times \frac{C^2}{4\pi^2}$$

$$\boxed{A = \frac{C^2}{4\pi}}$$

(c) The volume V of a cube as a function of the area A of its base.

Solution:

Let x be the length of each edge of a cube, then

$$V = x^3 \dots (i)$$

$$A = x^2 \Rightarrow \sqrt{A} = \sqrt{x^2}$$

$$\sqrt{A} = x$$

Put $x = \sqrt{A}$ in equation (i)

$$V = (\sqrt{A})^3$$

$$\boxed{V = A^{\frac{3}{2}}}$$

Q.4 Find the domain and the range of the function g defined below, and sketch the graph of g .

(i) $g(x) = 2x - 5$

Solution:

$$g(x) = 2x - 5$$

$$\text{Domain } g = R$$

$$\text{Range } g = R$$

(ii) $g(x) = \sqrt{x^2 - 4}$

Solution:

$$g(x) = \sqrt{x^2 - 4}$$

$g(x)$ is defined in real numbers if

$$x^2 - 4 \geq 0$$

$$x^2 \geq 4$$

$$\pm x \geq 2$$

$$x \leq -2 \text{ or } x \geq 2$$

$$\text{Domain } g = (-\infty, -2] \cup [2, \infty)$$

$$\text{Range } g = [0, \infty)$$

(iii) $g(x) = \sqrt{x+1}$

Solution:

$$g(x) = \sqrt{x+1}$$

$g(x)$ is defined in real numbers if

$$x+1 \geq 0$$

$$x \geq -1$$

$$\text{Domain } g = [-1, \infty)$$

$$\text{Range } g = [0, \infty)$$

(iv) $g(x) = |x-3|$

Solution:

$$g(x) = |x-3|$$

$$\text{Domain } g = R$$

$$\text{Range } g = [0, \infty)$$

(v) $g(x) = \begin{cases} 6x+7 & x \leq -2 \\ 4-3x & -2 < x \end{cases}$

Solution:

$$g(x) = \begin{cases} 6x+7 & x \leq -2 \\ 4-3x & -2 < x \end{cases}$$

$$\text{Domain } g = R$$

For range:

$$\text{If } x \leq -2$$

Multiplying both sides by 6

$$6x \leq -12$$

Adding 7 on both sides

$$6x+7 \leq -12+7$$

$$6x+7 \leq -5$$

$$g(x) \leq -5$$

$$g(x) \in (-\infty, -5]$$

$$\text{If } x > -2$$

Multiplying both sides by -3

$$-3x < 6$$

Adding 4 on both sides

$$4-3x < 4+6$$

$$4-3x < 10$$

$$g(x) < 10$$

$$g(x) \in (-\infty, 10)$$

$$g(x) \in (-\infty, -5] \cup (-\infty, 10)$$

$$\text{Range } g = (-\infty, 10)$$

(vi) $g(x) = \begin{cases} x-1 & , \quad x < 3 \\ 2x+1 & , \quad 3 \leq x \end{cases}$

Solution:

$$g(x) = \begin{cases} x-1 & , \quad x < 3 \\ 2x+1 & , \quad 3 \leq x \end{cases}$$

$$\text{Domain } g = R$$

For range:

$$\text{If } x < 3$$

Subtracting '1' on both sides

$$x-1 < 3-1$$

$$x-1 < 2$$

$$g(x) < 2$$

$$g(x) \in (-\infty, 2)$$

$$\text{If } x \geq 3$$

Multiplying both sides by 2

$$2x \geq 6$$

$$2x+1 \geq 7$$

$$g(x) \geq 7$$

$$g(x) \in [7, \infty)$$

$$\text{Range } g = (-\infty, 2) \cup [7, \infty)$$

(vii) $g(x) = \frac{x^2 - 3x + 2}{x+1}, \quad x \neq -1$

Solution:

$$g(x) = \frac{x^2 - 3x + 2}{x+1}, \quad x \neq -1$$

$g(x)$ is not defined if

$$x+1=0 \Rightarrow x=-1$$

$$\text{Domain } g = R - \{-1\}$$

Note: After the correction

For range:

$$g(x) = \frac{x^2 + 3x + 2}{x+1}, \quad x \neq -1$$

$$g(x) = \frac{x^2 + 2x + x + 2}{x+1}, \quad x \neq -1$$

$$g(x) = \frac{x(x+2)+1(x+2)}{x+1}, \quad x \neq -1$$

$$g(x) = \frac{(x+2)(x+1)}{x+1}, \quad x \neq -1$$

$$g(x) = x+2, \quad x \neq -1$$

By putting $x = -1$

$$g(-1) = -1 + 2 = 1$$

$$\text{Range } g = R - \{1\}$$

(viii) $g(x) = \frac{x^2 - 16}{x - 4}, \quad x \neq 4$

Solution:

$$g(x) = \frac{x^2 - 16}{x - 4}, \quad x \neq 4$$

$g(x)$ is not defined if

$$x - 4 = 0 \Rightarrow x = 4$$

$$\text{Domain } g = R - \{4\}$$

For range:

$$g(x) = \frac{(x-4)(x+4)}{x-4}, \quad x \neq 4$$

$$g(x) = x+4, \quad x \neq 4$$

By putting $x = 4$

$$g(4) = 4 + 4$$

$$g(4) = 8$$

$$\text{Range } g = R - \{8\}$$

Q.5 Given $f(x) = x^3 - ax^2 + bx + 1$.

If $f(2) = -3$ and $f(-1) = 0$.

Find the values of a and b .

Solution:

$$f(x) = x^3 - ax^2 + bx + 1$$

Putting $x = 2$ in $f(x)$

$$f(2) = (2)^3 - a(2)^2 + b(2) + 1$$

$$-3 = 8 - 4a + 2b + 1$$

$$-3 = 9 - 4a + 2b$$

$$-3 - 9 = -4a + 2b$$

$$-12 = -4a + 2b$$

$$-12 = -2(2a - b)$$

$$\frac{-12}{-2} = 2a - b$$

$$6 = 2a - b$$

$$2a - b = 6 \dots (i)$$

Putting $x = -1$ in $f(x)$

$$f(-1) = (-1)^3 - a(-1)^2 + b(-1) + 1$$

$$0 = -1 - a - b + 1$$

$$a + b = 0 \dots (ii)$$

Adding equation (i) and equation (ii)

$$2a - b = 6$$

$$a + b = 0$$

$$3a = 6$$

$$a = \frac{6}{3}$$

$$\boxed{a = 2}$$

Put $a = 2$ in equation (ii)

$$2 + b = 0$$

$$\boxed{b = -2}$$

Q.6 A stone falls from a height of 60m on the ground, the height h after x second is approximately given by $h(x) = 40 - 10x^2$.

(i) What is the height of the stone when:

(a) $x = 1$ sec.

Solution:

$$h(x) = 40 - 10x^2$$

Put $x = 1$ in $h(x)$

$$h(1) = 40 - 10(1)^2$$

$$= 40 - 10$$

$$= 30m$$

(b) $x = 1.5$ sec.

Solution:

$$h(x) = 40 - 10x^2$$

Put $x = 1.5$ in $h(x)$

$$h(1.5) = 40 - 10(1.5)^2$$

$$= 40 - 22.5$$

$$= 17.5m$$

(ii) When does the stone strike the ground?

Solution:

When stone strikes the ground,

then $h(x) = 0$

$$40 - 10x^2 = 0$$

$$40 = 10x^2$$

$$\frac{40}{10} = x^2$$

$$4 = x^2$$

By taking square root on both sides

$$x = \pm 2$$

As time is always a positive quantity, therefore,

$$x = 2 \text{ sec.}$$

Q.7 Show that the parametric equations:

(i) $x = at^2, y = 2at$ represent the equation of parabola $y^2 = 4ax$

Solution:

$$x = at^2 \dots (i)$$

$$y = 2at \Rightarrow t = \frac{y}{2a}$$

Put $t = \frac{y}{2a}$ in equation (i)

$$x = a \left(\frac{y}{2a} \right)^2$$

$$x = a \times \frac{y^2}{4a^2}$$

$$x = \frac{y^2}{4a}$$

$$\boxed{y^2 = 4ax}$$

(ii) $x = a \cos \theta, y = b \sin \theta$ represent the

equation of ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Solution:

$$x = a \cos \theta \Rightarrow \frac{x}{a} = \cos \theta$$

$$y = b \sin \theta \Rightarrow \frac{y}{b} = \sin \theta$$

Squaring and adding

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \cos^2 \theta + \sin^2 \theta$$

$$\boxed{\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1}$$

(iii) $x = a \sec \theta, y = b \tan \theta$ represent the equation of hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Solution:

$$x = a \sec \theta \Rightarrow \frac{x}{a} = \sec \theta$$

$$y = b \tan \theta \Rightarrow \frac{y}{b} = \tan \theta$$

Squaring and subtracting

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = \sec^2 \theta - \tan^2 \theta$$

$$\boxed{\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1}$$

Q.8 Prove the identities:

(i) $\sinh 2x = 2 \sinh x \cosh x$

Solution:

$$\text{L.H.S} = \sinh 2x$$

$$= \frac{e^{2x} - e^{-2x}}{2} \dots (i)$$

$$\text{R.H.S} = 2 \sinh x \cosh x$$

$$= 2 \left(\frac{e^x - e^{-x}}{2} \right) \left(\frac{e^x + e^{-x}}{2} \right)$$

$$= \frac{(e^x)^2 - (e^{-x})^2}{2}$$

$$= \frac{e^{2x} - e^{-2x}}{2} \dots (ii)$$

From equation (i) and equation (ii)

$$\boxed{\sinh 2x = 2 \sinh x \cosh x}$$

(ii) $\text{sech}^2 x = 1 - \tanh^2 x$

Solution:

$$\text{L.H.S} = \text{sech}^2 x$$

$$= \left(\frac{2}{e^x + e^{-x}} \right)^2$$

$$= \frac{4}{(e^x + e^{-x})^2} \dots (i)$$

$$\text{R.H.S} = 1 - \tanh^2 x$$

$$= 1 - \left(\frac{e^x - e^{-x}}{e^x + e^{-x}} \right)^2$$

$$\begin{aligned}
&= 1 - \frac{(e^x - e^{-x})^2}{(e^x + e^{-x})^2} \\
&= \frac{(e^x + e^{-x})^2 - (e^x - e^{-x})^2}{(e^x + e^{-x})^2} \\
&= \frac{e^{2x} + e^{-2x} + 2e^x \cdot e^{-x} - (e^{2x} + e^{-2x} - 2e^x \cdot e^{-x})}{(e^x + e^{-x})^2} \\
&= \frac{e^{2x} + e^{-2x} + 2 - e^{2x} - e^{-2x} + 2}{(e^x + e^{-x})^2} \\
&= \frac{4}{(e^x + e^{-x})^2} \dots (ii)
\end{aligned}$$

From equation (i) and equation (ii)

$$\boxed{\operatorname{sech}^2 x = 1 - \tanh^2 x}$$

(iii) $\operatorname{cosech}^2 x = \coth^2 x - 1$

Solution:

$$\begin{aligned}
\text{L.H.S} &= \operatorname{cosech}^2 x \\
&= \left(\frac{2}{e^x - e^{-x}} \right)^2 \\
&= \frac{4}{(e^x - e^{-x})^2} \dots (i) \\
\text{R.H.S} &= \coth^2 x - 1 \\
&= \left(\frac{e^x + e^{-x}}{e^x - e^{-x}} \right)^2 - 1 \\
&= \frac{(e^x + e^{-x})^2}{(e^x - e^{-x})^2} - 1 \\
&= \frac{(e^x + e^{-x})^2 - (e^x - e^{-x})^2}{(e^x - e^{-x})^2} \\
&= \frac{e^{2x} + e^{-2x} + 2e^x \cdot e^{-x} - (e^{2x} + e^{-2x} - 2e^x \cdot e^{-x})}{(e^x - e^{-x})^2} \\
&= \frac{e^{2x} + e^{-2x} + 2 - e^{2x} - e^{-2x} + 2}{(e^x - e^{-x})^2}
\end{aligned}$$

$$= \frac{4}{(e^x - e^{-x})^2} \dots (ii)$$

From equation (i) and equation (ii)

$$\boxed{\operatorname{cosech}^2 x = \coth^2 x - 1}$$

Q.9 Determine whether the given function f is even or odd.

(i) $f(x) = x^3 + x$

Solution:

$$f(x) = x^3 + x$$

Replace x by $-x$

$$\begin{aligned}
f(-x) &= (-x)^3 + (-x) \\
&= -x^3 - x \\
&= -(x^3 + x)
\end{aligned}$$

$$f(-x) = -f(x)$$

Hence $f(x)$ is an odd function.

(ii) $f(x) = (x+2)^2$

Solution:

$$f(x) = (x+2)^2$$

Replace x by $-x$

$$\begin{aligned}
f(-x) &= (-x+2)^2 \\
&= [-(x-2)]^2 \\
&= (x-2)^2
\end{aligned}$$

As neither $f(-x) = f(x)$ nor

$$f(-x) = -f(x)$$

Hence, $f(x)$ is neither even nor odd.

(iii) $f(x) = x\sqrt{x^2 + 5}$

Solution:

$$f(x) = x\sqrt{x^2 + 5}$$

Replace x by $-x$

$$\begin{aligned}
f(-x) &= -x\sqrt{(-x)^2 + 5} \\
f(-x) &= -x\sqrt{x^2 + 5} \\
f(-x) &= -f(x)
\end{aligned}$$

Hence $f(x)$ is an odd function.

(iv) $f(x) = \frac{x-1}{x+1}$, $x \neq -1$

Solution:

$$f(x) = \frac{x-1}{x+1}$$

Replace x by $-x$

$$\begin{aligned} f(-x) &= \frac{-x-1}{-x+1} \\ &= \frac{-(x+1)}{-(x-1)} \end{aligned}$$

$$f(-x) = \frac{x+1}{x-1}$$

As neither $f(-x) = f(x)$ nor

$$f(-x) = -f(x)$$

Hence, $f(x)$ is neither even nor odd.

(v) $f(x) = x^{\frac{2}{3}} + 6$

Solution:

$$f(x) = x^{\frac{2}{3}} + 6$$

Replace x by $-x$

$$\begin{aligned} f(-x) &= (-x)^{\frac{2}{3}} + 6 \\ &= \left[(-x)^2\right]^{\frac{1}{3}} + 6 \end{aligned}$$

$$= (x^2)^{\frac{1}{3}} + 6$$

$$= x^{\frac{2}{3}} + 6$$

$$f(-x) = f(x)$$

Hence $f(x)$ is even function.

(vi) $f(x) = \frac{x^3 - x}{x^2 + 1}$

Solution:

$$f(x) = \frac{x^3 - x}{x^2 + 1}$$

Replace x by $-x$

$$f(-x) = \frac{(-x)^3 - (-x)}{(-x)^2 + 1}$$

$$= \frac{-x^3 + x}{x^2 + 1}$$

$$= \frac{-(x^3 - x)}{x^2 + 1}$$

$$= -\frac{x^3 - x}{x^2 + 1}$$

$$f(-x) = -f(x)$$

Hence, $f(x)$ is an odd function.

Composition of Functions:

Let f be a function from set X to set Y and g be a function from set Y to set Z .

The composition of f and g is a function, denoted by gof , from X to Z and is defined by

$$(gof)(x) = g(f(x)) = gf(x), \quad \forall x \in X$$

Example 1: Let the real valued functions f and g be defined by $f(x) = 2x+1$ and

$$g(x) = x^2 - 1.$$

Obtain the expressions for (i) $fg(x)$ (ii) $gf(x)$ (iii) $f^2(x)$ (iv) $g^2(x)$

Solution:

(i) $fg(x) = f(g(x)) = f(x^2 - 1) = 2(x^2 - 1) + 1 = 2x^2 - 1$

(ii) $gf(x) = g(f(x)) = g(2x+1) = (2x+1)^2 - 1 = 4x^2 + 4x$

(iii) $f^2(x) = f(f(x)) = f(2x+1) = 2(2x+1) + 1 = 4x+3$

(iv) $g^2(x) = g(g(x)) = g(x^2 - 1) = (x^2 - 1)^2 - 1 = x^4 - 2x^2$

We observe from (i) and (ii) that $fg(x) \neq gf(x)$

Note:

- (i) It is important to note that in general, $gf(x) \neq fg(x)$, because $gf(x)$ means that f is applied first then followed by g , whereas $fg(x)$ means that g is applied first then followed by f .
- (ii) We usually write ff as f^2 and fff as f^3 and so on.
- (iii) $f^n(x) \neq [f(x)]^n, n \in \mathbb{Z}^+ \cup \{0\}$

Inverse of a Function:

Let f be a one-one function from X onto Y . The **inverse function** of f , denoted by f^{-1} , is a function from Y onto X and is defined by $x = f^{-1}(y), \forall y \in Y$ iff $y = f(x), \forall x \in X$.

Example 2: Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function defined by $f(x) = 2x + 1$. find $f^{-1}(x)$

Solution:

We find the inverse of f as follows:

$$\text{Write } f(x) = 2x + 1 = y$$

So that y is the image of x under f .

Now solve this equation for x as follows:

$$\begin{aligned} y &= 2x + 1 \\ \Rightarrow 2x &= y - 1 \\ \Rightarrow x &= \frac{y - 1}{2} \\ \therefore f^{-1}(y) &= \frac{1}{2}(y - 1) \quad [\because x = f^{-1}(y)] \end{aligned}$$

To find $f^{-1}(x)$, replace y by x .

$$\therefore \boxed{f^{-1}(x) = \frac{1}{2}(x - 1)}$$

Example 3: Without finding the inverse, state the domain and range of f^{-1} , where

$$f(x) = 2 + \sqrt{x - 1}$$

Solution:

We see that f is not defined when $x < 1$.

$$\therefore \text{Domain } f = [1, +\infty)$$

As x varies over the interval $[1, +\infty)$, the value of $\sqrt{x - 1}$ varies over the interval $[0, +\infty)$. So the value of $f(x) = 2 + \sqrt{x - 1}$ varies over the interval $[2, +\infty)$.

$$\text{Therefore range } f = [2, +\infty)$$

By definition of inverse function f^{-1} , we have

$$\text{Domain } f^{-1} = \text{range } f = [2, +\infty)$$

$$\text{Range } f^{-1} = \text{domain } f = [1, +\infty).$$

EXERCISE 1.2

Q.1 The real valued functions f and g are defined below. Find

(a) $f \circ g(x)$

(b) $g \circ f(x)$

(c) $f \circ f(x)$

(d) $g \circ g(x)$

(i) $f(x) = 2x + 1, \quad g(x) = \frac{3}{x-1}, \quad x \neq 1$

Solution:

$$f(x) = 2x + 1, \quad g(x) = \frac{3}{x-1}, \quad x \neq 1$$

(a) $f \circ g(x)$

$$\begin{aligned} f \circ g(x) &= f(g(x)) \\ &= f\left(\frac{3}{x-1}\right) \end{aligned}$$

$$\text{Replace } x \text{ by } \frac{3}{x-1} \text{ in } f(x).$$

$$\begin{aligned} &= 2\left(\frac{3}{x-1}\right) + 1 \\ &= \frac{6}{x-1} + 1 \\ &= \frac{6+x-1}{x-1} \end{aligned}$$

$$\boxed{f \circ g(x) = \frac{x+5}{x-1}}$$

(b) $g \circ f(x)$

$$\begin{aligned} g \circ f(x) &= g(f(x)) \\ &= g(2x+1) \end{aligned}$$

$$\text{Replace } x \text{ by } 2x+1 \text{ in } g(x)$$

$$= \frac{3}{2x+1-1}$$

$$\boxed{g \circ f(x) = \frac{3}{2x}}$$

(c) $f \circ f(x)$

$$\begin{aligned} f \circ f(x) &= f(f(x)) \\ &= f(2x+1) \end{aligned}$$

$$\text{Replace } x \text{ by } 2x+1 \text{ in } f(x)$$

$$= 2(2x+1) + 1$$

$$= 4x + 2 + 1$$

$$\boxed{f \circ f(x) = 4x + 3}$$

(d) $g \circ g(x)$

$$g \circ g(x) = g(g(x))$$

$$= g\left(\frac{3}{x-1}\right)$$

$$\text{Replace } x \text{ by } \frac{3}{x-1} \text{ in } g(x)$$

$$\begin{aligned} &= \frac{3}{\frac{3}{x-1} - 1} \\ &= \frac{3}{\frac{3-(x-1)}{x-1}} \\ &= \frac{3(x-1)}{3-x+1} \end{aligned}$$

$$\boxed{g \circ g(x) = \frac{3(x-1)}{4-x}}$$

(ii) $f(x) = \sqrt{x+1}, \quad g(x) = \frac{1}{x^2}, \quad x \neq 0$

(a) $f \circ g(x)$

Solution:

$$f \circ g(x) = f(g(x))$$

$$= f\left(\frac{1}{x^2}\right)$$

$$\text{Replace } x \text{ by } \frac{1}{x^2} \text{ in } f(x)$$

$$= \sqrt{\frac{1}{x^2} + 1}$$

$$= \sqrt{\frac{1+x^2}{x^2}}$$

$$\boxed{f \circ g(x) = \frac{\sqrt{1+x^2}}{x}}$$

(b) $gof(x)$ **Solution:**

$$gof(x) = g(f(x))$$

$$= g(\sqrt{x+1})$$

Replace x by $\sqrt{x+1}$ in $g(x)$

$$= \frac{1}{(\sqrt{x+1})^2}$$

$$\boxed{gof(x) = \frac{1}{x+1}}$$

(c) $fof(x)$ **Solution:**

$$fof(x) = f(f(x))$$

$$= f(\sqrt{x+1})$$

Replace x by $\sqrt{x+1}$ in $f(x)$

$$\boxed{fof(x) = \sqrt{\sqrt{x+1}+1}}$$

(d) $gog(x)$ **Solution:**

$$gog(x) = g(g(x))$$

$$= g\left(\frac{1}{x^2}\right)$$

Replace x by $\frac{1}{x^2}$ in $g(x)$

$$= \frac{1}{\left(\frac{1}{x^2}\right)^2}$$

$$= \frac{1}{\left(\frac{1}{x^4}\right)}$$

$$\boxed{gog(x) = x^4}$$

(iii) $f(x) = \frac{1}{\sqrt{x-1}}, x \neq 1, g(x) = (x^2+1)^2$ (a) $fog(x)$ **Solution:**

$$fog(x) = f(g(x))$$

$$= f\left((x^2+1)^2\right)$$

Replace x by $(x^2+1)^2$ in $f(x)$

$$= \frac{1}{\sqrt{(x^2+1)^2-1}}$$

$$= \frac{1}{\sqrt{x^4+2x^2+1-1}}$$

$$= \frac{1}{\sqrt{x^4+2x^2}}$$

$$= \frac{1}{\sqrt{x^2(x^2+2)}}$$

$$\boxed{fog(x) = \frac{1}{x\sqrt{x^2+2}}}$$

(b) $gof(x)$ **Solution:**

$$gof(x) = g(f(x))$$

$$= g\left(\frac{1}{\sqrt{x-1}}\right)$$

Replace x by $\frac{1}{\sqrt{x-1}}$ in $g(x)$.

$$= \left[\left(\frac{1}{\sqrt{x-1}}\right)^2 + 1\right]^2$$

$$= \left(\frac{1}{x-1} + 1\right)^2$$

$$= \left(\frac{1+x-1}{x-1}\right)^2$$

$$= \left(\frac{x}{x-1}\right)^2$$

$$\boxed{gof(x) = \frac{x^2}{(x-1)^2}}$$

(c) $fof(x)$ **Solution:**

$$fof(x) = f(f(x))$$

$$= f\left(\frac{1}{\sqrt{x-1}}\right)$$

Replace x by $\frac{1}{\sqrt{x-1}}$ in $f(x)$

$$= \frac{1}{\sqrt{\sqrt{x-1}-1}}$$

$$= \frac{1}{\sqrt{\frac{1-\sqrt{x-1}}{\sqrt{x-1}}}}$$

$$\boxed{f \circ f(x) = \sqrt{\frac{\sqrt{x-1}}{1-\sqrt{x-1}}}}$$

(d) $g \circ g(x)$ **Solution:**

$$g \circ g(x) = g(g(x))$$

$$= g((x^2+1)^2)$$

Replace x by $(x^2+1)^2$ in $g(x)$

$$= \left[\left((x^2+1)^2 \right)^2 + 1 \right]^2$$

$$\boxed{g \circ g(x) = \left((x^4 + 2x^2 + 1)^2 + 1 \right)^2}$$

(iv) $f(x) = 3x^4 - 2x^2$, $g(x) = \frac{2}{\sqrt{x}}$, $x \neq 0$ (a) $f \circ g(x)$ **Solution:**

$$f \circ g(x) = f(g(x))$$

$$= f\left(\frac{2}{\sqrt{x}}\right)$$

Replace x by $\frac{2}{\sqrt{x}}$ in $f(x)$

$$f \circ g(x) = 3\left(\frac{2}{\sqrt{x}}\right)^4 - 2\left(\frac{2}{\sqrt{x}}\right)^2$$

$$= 3\left(\frac{16}{x^2}\right) - 2\left(\frac{4}{x}\right)$$

$$= \frac{48}{x^2} - \frac{8}{x}$$

$$= \frac{48-8x}{x^2}$$

$$\boxed{f \circ g(x) = \frac{8(6-x)}{x^2}}$$

(b) $g \circ f(x)$ **Solution:**

$$g \circ f(x) = g(f(x))$$

$$= g(3x^4 - 2x^2)$$

Replace x by $3x^4 - 2x^2$ in $g(x)$

$$= \frac{2}{\sqrt{3x^4 - 2x^2}}$$

$$= \frac{2}{\sqrt{x^2(3x^2 - 2)}}$$

$$\boxed{g \circ f(x) = \frac{2}{x\sqrt{3x^2 - 2}}}$$

(c) $f \circ f(x)$ **Solution:**

$$f \circ f(x) = f(f(x))$$

$$= f(3x^4 - 2x^2)$$

Replace x by $3x^4 - 2x^2$ in $f(x)$

$$\boxed{f \circ f(x) = 3(3x^4 - 2x^2)^4 - 2(3x^4 - 2x^2)^2}$$

(d) $g \circ g(x)$ **Solution:**

$$g \circ g(x) = g(g(x))$$

$$= g\left(\frac{2}{\sqrt{x}}\right)$$

Replace x by $\frac{2}{\sqrt{x}}$ in $g(x)$

$$= \frac{2}{\sqrt{\frac{2}{\sqrt{x}}}}$$

$$= \frac{2}{\frac{\sqrt{2}}{\sqrt{\sqrt{x}}}}$$

$$= \frac{2\sqrt{\sqrt{x}}}{\sqrt{2}}$$

$$= \sqrt{2}\sqrt{\sqrt{x}}$$

$$\boxed{g \circ g(x) = \sqrt{2}\sqrt{\sqrt{x}}}$$

Q.2 For the real valued function f defined below find

(a) $f^{-1}(x)$

(b) $f^{-1}(-1)$ and verify

$$f(f^{-1}(x)) = f^{-1}(f(x)) = x$$

(i) $f(x) = -2x + 8$

(a) $f^{-1}(x)$

Solution:

$$y = f(x) = -2x + 8$$

$$y = -2x + 8$$

$$2x = 8 - y$$

$$x = \frac{8-y}{2}$$

$$\therefore y = f(x) \Rightarrow f^{-1}(y) = x$$

$$f^{-1}(y) = \frac{8-y}{2}$$

Replacing y by x

$$f^{-1}(x) = \frac{8-x}{2}$$

(b) $f^{-1}(-1)$

Solution:

$$f^{-1}(x) = \frac{8-x}{2}$$

Putting $x = -1$

$$f^{-1}(-1) = \frac{8-(-1)}{2}$$

$$f^{-1}(-1) = \frac{9}{2}$$

Verification:

$$f(f^{-1}(x)) = f\left(\frac{8-x}{2}\right)$$

$$\text{Replace } x \text{ by } \frac{8-x}{2} \text{ in } f(x)$$

$$= -2\left(\frac{8-x}{2}\right) + 8$$

$$= -8 + x + 8$$

$$= x$$

$$\text{Now } f^{-1}(f(x)) = f^{-1}(-2x+8)$$

$$\text{Replace } x \text{ by } -2x+8 \text{ in } f^{-1}(x)$$

$$= \frac{8-(-2x+8)}{2}$$

$$= \frac{8+2x-8}{2}$$

$$= \frac{2x}{2}$$

$$= x$$

$$f(f^{-1}(x)) = f^{-1}(f(x)) = x$$

(ii) $f(x) = 3x^3 + 7$

(a) $f^{-1}(x)$

Solution:

$$y = f(x) = 3x^3 + 7$$

$$y = 3x^3 + 7$$

$$y - 7 = 3x^3$$

$$\frac{y-7}{3} = x^3$$

By taking cube root on both sides.

$$\left(\frac{y-7}{3}\right)^{\frac{1}{3}} = x$$

$$\therefore y = f(x) \Rightarrow f^{-1}(y) = x$$

$$f^{-1}(y) = \left(\frac{y-7}{3}\right)^{\frac{1}{3}}$$

Replacing y by x

$$f^{-1}(x) = \left(\frac{x-7}{3}\right)^{\frac{1}{3}}$$

(b) $f^{-1}(-1)$

Solution:

$$f^{-1}(x) = \left(\frac{x-7}{3}\right)^{\frac{1}{3}}$$

By putting $x = -1$

$$f^{-1}(-1) = \left(\frac{-1-7}{3}\right)^{\frac{1}{3}}$$

$$= \left(\frac{-8}{3}\right)^{\frac{1}{3}}$$

Verification:

$$f(f^{-1}(x)) = f\left(\left(\frac{x-7}{3}\right)^{\frac{1}{3}}\right)$$

Replace x by $\left(\frac{x-7}{3}\right)^{\frac{1}{3}}$ in $f(x)$

$$= 3\left[\left(\frac{x-7}{3}\right)^{\frac{1}{3}}\right]^3 + 7$$

$$= 3\left(\frac{x-7}{3}\right) + 7$$

$$= x - 7 + 7$$

$$= x$$

Now $f^{-1}(f(x)) = f^{-1}(3x^3 + 7)$

Replace x by $3x^3 + 7$ in $f^{-1}(x)$

$$= \left(\frac{3x^3 + 7 - 7}{3}\right)^{\frac{1}{3}}$$

$$= \left(\frac{3x^3}{3}\right)^{\frac{1}{3}}$$

$$= (x^3)^{\frac{1}{3}}$$

$$= x$$

$$\boxed{f(f^{-1}(x)) = f^{-1}(f(x)) = x}$$

(iii) $f(x) = (-x+9)^3$

(a) $f^{-1}(x)$

Solution:

$$y = f(x) = (-x+9)^3$$

$$y = (-x+9)^3$$

By taking cube root on both sides.

$$y^{\frac{1}{3}} = [(-x+9)^3]^{\frac{1}{3}}$$

$$y^{\frac{1}{3}} = -x+9$$

$$x = 9 - y^{\frac{1}{3}}$$

$$\therefore y = f(x) \Rightarrow f^{-1}(y) = x$$

$$f^{-1}(y) = 9 - y^{\frac{1}{3}}$$

Replacing y by x

$$\boxed{f^{-1}(x) = 9 - x^{\frac{1}{3}}}$$

(b) $f^{-1}(-1)$

Solution:

$$f^{-1}(x) = 9 - x^{\frac{1}{3}}$$

Putting $x = -1$

$$f^{-1}(-1) = 9 - (-1)^{\frac{1}{3}}$$

Verification:

$$f(f^{-1}(x)) = f\left(9 - x^{\frac{1}{3}}\right)$$

Replace x by $9 - x^{\frac{1}{3}}$ in $f(x)$

$$= \left(-\left(9 - x^{\frac{1}{3}}\right) + 9\right)^3$$

$$= \left(-9 + x^{\frac{1}{3}} + 9\right)^3$$

$$= \left(x^{\frac{1}{3}}\right)^3$$

$$= x$$

Now $f^{-1}(f(x)) = f^{-1}((-x+9)^3)$

Replace x by $(-x+9)^3$ in $f(x)$

$$= 9 - \left((-x+9)^3\right)^{\frac{1}{3}}$$

$$= 9 - (-x+9)$$

$$= 9 + x - 9$$

$$= x$$

$$\boxed{f(f^{-1}(x)) = f^{-1}(f(x)) = x}$$

(iv) $f(x) = \frac{2x+1}{x-1}, x > 1$

(a) $f^{-1}(x)$

Solution:

$$y = f(x) = \frac{2x+1}{x-1}$$

$$y = \frac{2x+1}{x-1}$$

$$y(x-1) = 2x+1$$

$$xy - y = 2x+1$$

$$xy - 2x = y+1$$

$$x(y-2) = y+1$$

$$x = \frac{y+1}{y-2}$$

$$\therefore y = f(x) \Rightarrow f^{-1}(y) = x$$

$$f^{-1}(y) = \frac{y+1}{y-2}$$

Replacing y by x

$$\boxed{f^{-1}(x) = \frac{x+1}{x-2}}$$

(b) $f^{-1}(-1)$

Solution:

$$f^{-1}(x) = \frac{x+1}{x-2}$$

Putting $x = -1$

$$f^{-1}(-1) = \frac{-1+1}{-1-2}$$

$$f^{-1}(-1) = 0$$

Verification:

$$f(f^{-1}(x)) = f\left(\frac{x+1}{x-2}\right)$$

Replace x by $\frac{x+1}{x-2}$ in $f(x)$

$$= \frac{2\left(\frac{x+1}{x-2}\right)+1}{\frac{x+1}{x-2}-1}$$

$$\frac{2(x+1)+x-2}{x-2}$$

$$= \frac{x-2}{(x+1)-(x-2)}$$

$$= \frac{x-2}{x-2}$$

$$= \frac{2x+2+x-2}{x+1-x+2}$$

$$= \frac{3x}{3}$$

$$= x$$

$$\text{Now } f^{-1}(f(x)) = f^{-1}\left(\frac{2x+1}{x-1}\right)$$

Replace x by $\frac{2x+1}{x-1}$ in $f^{-1}(x)$

$$f^{-1}\left(\frac{2x+1}{x-1}\right) = \frac{\frac{2x+1}{x-1}+1}{\frac{2x+1}{x-1}-2}$$

$$= \frac{\frac{2x+1+x-1}{x-1}}{\frac{2x+1-2(x-1)}{x-1}}$$

$$= \frac{2x+1+x-1}{(2x+1)-2(x-1)}$$

$$= \frac{3x}{2x+1-2x+2}$$

$$= \frac{3x}{3}$$

$$= x$$

$$\boxed{f(f^{-1}(x)) = f^{-1}(f(x)) = x}$$

Q.3 Without finding the inverse, state the domain and range of f^{-1} .

(i) $f(x) = \sqrt{x+2}$

Solution:

$$\text{As } f(x) = \sqrt{x+2}$$

$f(x)$ is defined on real numbers if

$$x+2 \geq 0$$

$$x \geq -2$$

$$\text{Domain } f = [-2, \infty)$$

$$\text{Range } f = [0, \infty)$$

By the definition of inverse function

f^{-1} , we have

$$\text{Domain } f^{-1} = \text{Range } f = [0, \infty)$$

$$\text{Range } f^{-1} = \text{Domain } f = [-2, \infty)$$

(ii) $f(x) = \frac{x-1}{x-4}, x \neq 4$

Solution:

$$f(x) = \frac{x-1}{x-4}, x \neq 4$$

$$\text{Domain } f = R - \{4\}$$

$$\text{Range } f = R - \{1\}$$

By the definition of inverse function f^{-1} , we have

$$\text{Domain } f^{-1} = \text{Range } f = R - \{1\}$$

$$\text{Range } f^{-1} = \text{Domain } f = R - \{4\}$$

(iii) $f(x) = \frac{1}{x+3}, x \neq -3$

Solution:

$$f(x) = \frac{1}{x+3}$$

$$\text{Domain } f = R - \{-3\}$$

$$\text{Range } f = R - \{0\}$$

By the definition of inverse function f^{-1} , we have

$$\text{Domain } f^{-1} = \text{Range } f = R - \{0\}$$

$$\text{Range } f^{-1} = \text{Domain } f = R - \{-3\}$$

(iv) $f(x) = (x-5)^2, x \geq 5$

Solution:

$$f(x) = (x-5)^2, x \geq 5$$

$$\text{Domain } f = [5, \infty)$$

$$\text{Range } f = [0, \infty)$$

By the definition of inverse function f^{-1} , we have

$$\text{Domain } f^{-1} = \text{Range } f = [0, \infty)$$

$$\text{Range } f^{-1} = \text{Domain } f = [5, \infty)$$

Limit of a Function:

Let a function $f(x)$ be defined in an open interval near the number a (need not at a).

If, as x approaches a from both left and right side of a , $f(x)$ approaches a specific number L , then L is called the limit of $f(x)$ as x approaches a . Symbolically it is written as: $\lim_{x \rightarrow a} f(x) = L$ (read as “limit of $f(x)$, as $x \rightarrow a$, is L ”)

Example: If $P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$ is a polynomial function of degree n , then show that: $\lim_{x \rightarrow c} P(x) = P(c)$

Solution:

Using the theorems on limits, we have

$$\begin{aligned} \lim_{x \rightarrow c} P(x) &= \lim_{x \rightarrow c} (a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0) \\ &= a_n \lim_{x \rightarrow c} x^n + a_{n-1} \lim_{x \rightarrow c} x^{n-1} + \dots + a_1 \lim_{x \rightarrow c} x + \lim_{x \rightarrow c} a_0 \\ &= a_n c^n + a_{n-1} c^{n-1} + \dots + a_1 c + a_0 \end{aligned}$$

$$\therefore \lim_{x \rightarrow c} P(x) = P(c)$$

Limits Of Important Functions:

Theorem: Prove that $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1}$, where n is an integer and $a > 0$.

Proof: Case-I: Suppose n is a positive integer.

By substituting $x = a$, we get $\left(\frac{0}{0}\right)$ form, so we make factors as follows:

$$x^n - a^n = (x - a)(x^{n-1} + ax^{n-2} + a^2x^{n-3} + \dots + a^{n-1})$$

$$\begin{aligned}
 \lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} &= \lim_{x \rightarrow a} \frac{(x-a)(x^{n-1} + ax^{n-2} + a^2x^{n-3} + \dots + a^{n-1})}{(x-a)} \\
 &= \lim_{x \rightarrow a} (x^{n-1} + ax^{n-2} + a^2x^{n-3} + \dots + a^{n-1}) \\
 &= a^{n-1} + aa^{n-2} + a^2a^{n-3} + \dots + a^{n-1} \\
 &= a^{n-1} + a^{n-1} + a^{n-1} + \dots + a^{n-1} \quad (n \text{ terms}) \\
 &= n a^{n-1}
 \end{aligned}$$

Case-II: Suppose n is a negative integer (say $n = -m$), where m is a positive integer

$$\begin{aligned}
 \text{Now } \frac{x^n - a^n}{x - a} &= \frac{x^{-m} - a^{-m}}{x - a} \\
 &= \frac{\frac{1}{x^m} - \frac{1}{a^m}}{x - a} \\
 &= \frac{\frac{a^m - x^m}{x^m a^m}}{x - a} \\
 &= \frac{a^m - x^m}{x^m a^m (x - a)} \\
 &= \frac{-1}{x^m a^m} \left(\frac{x^m - a^m}{x - a} \right), \quad (a \neq 0)
 \end{aligned}$$

$$\begin{aligned}
 \text{Now, } \lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} &= \lim_{x \rightarrow a} \frac{-1}{x^m a^m} \left(\frac{x^m - a^m}{x - a} \right) \\
 &= \left(\lim_{x \rightarrow a} \frac{-1}{x^m a^m} \right) \times \left(\lim_{x \rightarrow a} \frac{x^m - a^m}{x - a} \right) \\
 &= \frac{-1}{a^m a^m} m a^{m-1} \quad (\text{By Case-I}) \\
 &= -m a^{m-1-m-m} \\
 &= (-m) a^{-m-1} \\
 &= n a^{n-1} \quad (n = -m)
 \end{aligned}$$

Example 1: Evaluate $\lim_{x \rightarrow -\infty} \frac{4x^4 - 5x^3}{3x^5 + 2x^2 + 1}$

Solution:

Since $x < 0$, so dividing up and down by $(-x)^5 = -x^5$, we get

$$\begin{aligned}
 \lim_{x \rightarrow -\infty} \frac{4x^4 - 5x^3}{3x^5 + 2x^2 + 1} &= \lim_{x \rightarrow -\infty} \frac{-\frac{4}{x} + \frac{5}{x^2}}{-3 - \frac{2}{x^3} - \frac{1}{x^5}} \\
 &= \frac{0+0}{-3-0-0} = 0
 \end{aligned}$$

Example 2: Evaluate $\lim_{x \rightarrow -\infty} \frac{2-3x}{\sqrt{3+4x^2}}$

Solution:

Here $\sqrt{x^2} = |x| = -x$ as $x < 0$

∴ Dividing up and down by $-x$, we get

$$\begin{aligned} \lim_{x \rightarrow -\infty} \frac{2-3x}{\sqrt{3+4x^2}} &= \lim_{x \rightarrow -\infty} \frac{-\frac{2}{x}+3}{\sqrt{\frac{3}{x^2}+4}} \\ &= \frac{0+3}{\sqrt{0+4}} = \frac{3}{2} \end{aligned}$$

Example 3: Evaluate $\lim_{x \rightarrow +\infty} \frac{2-3x}{\sqrt{3+4x^2}}$

Solution:

Here $\sqrt{x^2} = |x| = x$ as $x > 0$

∴ Dividing up and down by x , we get

$$\begin{aligned} \lim_{x \rightarrow +\infty} \frac{2-3x}{\sqrt{3+4x^2}} &= \lim_{x \rightarrow +\infty} \frac{\frac{2}{x}-3}{\sqrt{\frac{3}{x^2}+4}} \\ &= \frac{0-3}{\sqrt{0+4}} = -\frac{3}{2} \end{aligned}$$

Theorem: Prove that $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$

Proof:

By the binomial theorem, we have

$$\begin{aligned} \left(1 + \frac{1}{n}\right)^n &= 1 + n\left(\frac{1}{n}\right) + \frac{n(n-1)}{2!}\left(\frac{1}{n}\right)^2 + \frac{n(n-1)(n-2)}{3!}\left(\frac{1}{n}\right)^3 + \dots \\ &= 1 + 1 + \frac{1}{2!}n(n-1) \times \frac{1}{n^2} + \frac{1}{3!}n(n-1)(n-2) \times \frac{1}{n^3} + \dots \\ &= 2 + \frac{1}{2!}n^2\left(1 - \frac{1}{n}\right) \cdot \frac{1}{n^2} + \frac{1}{3!}n^3\left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right) \frac{1}{n^3} + \dots \\ &= 2 + \frac{1}{2!}\left(1 - \frac{1}{n}\right) + \frac{1}{3!}\left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right) + \dots \end{aligned}$$

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = \lim_{n \rightarrow \infty} \left[2 + \frac{1}{2!}\left(1 - \frac{1}{n}\right) + \frac{1}{3!}\left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right) + \dots \right]$$

As $n \rightarrow \infty$, $\frac{1}{n}, \frac{2}{n}, \frac{3}{n}, \dots$ all tend to zero.

$$\begin{aligned} &= 2 + \frac{1}{2!}(1-0) + \frac{1}{3!}(1-0)(1-0) + \dots \\ &= 2 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \dots \\ &= 2 + 0.5 + 0.166667 + 0.0416667 + \dots \\ &= 2.718281\dots \end{aligned}$$

As approximate value of e is 2.718281, so

$$\boxed{\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right) = e}$$

Deduction: $\lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = e$

Proof:

We know that $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$

Put $n = \frac{1}{x}$, then $\frac{1}{n} = x$

When $n \rightarrow \infty$, $x \rightarrow 0$

As $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$

$$\boxed{\lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = e}$$

Theorem: Prove that $\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log_e a$

Proof:

Put $a^x - 1 = y$ (i)

Then $a^x = 1 + y$

Taking logarithm on both sides with base a .

$$\log_a a^x = \log_a (1 + y)$$

$$x \cdot \log_a a = \log_a (1 + y)$$

$$\text{So } x = \log_a (1 + y)$$

From (i) when $x \rightarrow 0$, $y \rightarrow 0$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{a^x - 1}{x} &= \lim_{y \rightarrow 0} \frac{y}{\log_a (1 + y)} \\ &= \lim_{y \rightarrow 0} \frac{1}{\frac{1}{y} \log_a (1 + y)} \\ &= \lim_{y \rightarrow 0} \frac{1}{\log_a (1 + y)^{\frac{1}{y}}} \\ &= \frac{1}{\log_a \left(\lim_{y \rightarrow 0} (1 + y)^{\frac{1}{y}} \right)} \\ &= \frac{1}{\log_a e} \quad \left(\because \lim_{y \rightarrow 0} (1 + y)^{\frac{1}{y}} = e \right) \end{aligned}$$

$$\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log_e a$$

Deduction: $\lim_{x \rightarrow 0} \left(\frac{e^x - 1}{x} \right) = \log_e e = 1$

We know that $\lim_{x \rightarrow 0} \left(\frac{a^x - 1}{x} \right) = \log_e a$ (i)

Put $a = e$ in (i)

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = \log_e e = 1$$

The Sandwich Theorem:

Let f, g and h be functions such that $f(x) \leq g(x) \leq h(x)$ for all numbers x in some open interval containing c , except possibly at c itself.

If $\lim_{x \rightarrow c} f(x) = L$ and $\lim_{x \rightarrow c} h(x) = L$, then $\lim_{x \rightarrow c} g(x) = L$

Theorem: If θ is measured in radian, then $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$

Proof: Take θ a positive acute central angle of a circle with radius $r = 1$.

Produce $\overline{OB}$ to D , so that $\overline{AD} \perp \overline{OA}$.

Draw $\overline{BC} \perp \overline{OA}$. Join A and B . As show in figure, OAB represent a sector of the circle.

Given $|\overline{OA}| = |\overline{OB}| = 1$ (radii of unit circle)

In right $\triangle OCB$, $\sin \theta = \frac{|\overline{BC}|}{|\overline{OB}|} = |\overline{BC}|$ ($|\overline{OB}| = 1$)

In right $\triangle OAD$, $\tan \theta = \frac{|\overline{AD}|}{|\overline{OA}|} = |\overline{AD}|$ ($|\overline{OA}| = 1$)

In terms of θ , the areas are expressed as:

(i) Area of $\triangle OAB = \frac{1}{2} |\overline{OA}| |\overline{BC}| = \frac{1}{2} (1) \sin \theta = \frac{1}{2} \sin \theta$

(ii) Area of sector $OAB = \frac{1}{2} r^2 \theta = \frac{1}{2} (1)^2 \theta = \frac{1}{2} \theta$ ($r = 1$)

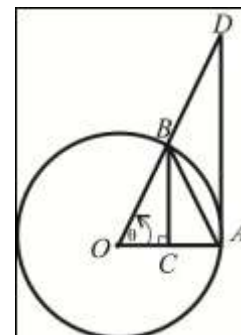
(iii) Area of $\triangle OAD = \frac{1}{2} |\overline{OA}| |\overline{AD}| = \frac{1}{2} (1) \tan \theta$

From the figure, we see that

Area of $\triangle OAB <$ Area of sector $OAB <$ Area of $\triangle OAD$

$$\frac{1}{2} \sin \theta < \frac{\theta}{2} < \frac{1}{2} \tan \theta$$

$$\frac{1}{2} \sin \theta < \frac{\theta}{2} < \frac{1}{2} \frac{\sin \theta}{\cos \theta}$$



As $\sin \theta$ is positive, so on division by $\frac{1}{2} \sin \theta$, we get

$$\frac{\frac{1}{2} \sin \theta}{\frac{1}{2} \sin \theta} < \frac{\frac{\theta}{2}}{\frac{1}{2} \sin \theta} < \frac{\frac{1}{2} \sin \theta}{\frac{1}{2} \sin \theta}$$

$$1 < \frac{\theta}{\sin \theta} < \frac{1}{\cos \theta} \quad \left(\because 0 < \theta < \frac{\pi}{2} \right)$$

$$\text{i.e., } 1 > \frac{\sin \theta}{\theta} > \cos \theta \quad \text{or} \quad \cos \theta < \frac{\sin \theta}{\theta} < 1$$

when $\theta \rightarrow 0$, $\cos \theta \rightarrow 1$

since $\frac{\sin \theta}{\theta}$ is sandwiched between 1 and a quantity approaching 1 itself.

So, by the sandwich theorem, it must also approach 1.

$$\text{i.e., } \boxed{\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1}$$

Limits of Important Functions:

$\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1}$, n is an integer, $a > 0$	
$\lim_{x \rightarrow a} \frac{x^n - a^n}{x^m - a^m} = \frac{n}{m} a^{n-m}$	
$\lim_{x \rightarrow \pm\infty} \frac{1}{x} = 0$, $x \neq 0$	
$\lim_{x \rightarrow \pm\infty} \frac{a}{x^p} = 0$ where $p \in Q^+$, $a \in R$	
$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$	$\lim_{x \rightarrow +\infty} (e^x) = \infty$
$\lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = e$	$\lim_{x \rightarrow -\infty} (e^x) = \lim_{x \rightarrow -\infty} \left(\frac{1}{e^{-x}}\right) = 0$
$\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \log_e a$	$\lim_{x \rightarrow \pm\infty} \left(\frac{a}{x}\right) = 0$, $a \in R$
$\lim_{x \rightarrow 0} \left(\frac{e^x - 1}{x}\right) = \log_e e = 1$	$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$, where θ is in radians.

EXERCISE 1.3

Q.1 Evaluate each limit by using theorems of limits:

(i) $\lim_{x \rightarrow 3} (2x + 4)$

Solution:

$$\lim_{x \rightarrow 3} (2x + 4)$$

$$\because \lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$$

$$= \lim_{x \rightarrow 3} (2x) + \lim_{x \rightarrow 3} (4)$$

$$\because \lim_{x \rightarrow a} [kf(x)] = k \left[\lim_{x \rightarrow a} f(x) \right]$$

$$= 2 \lim_{x \rightarrow 3} (x) + 4$$

$$= 2(3) + 4$$

$$= 6 + 4$$

$$= 10$$

$$\boxed{\lim_{x \rightarrow 3} (2x + 4) = 10}$$

(ii) $\lim_{x \rightarrow 1} (3x^2 - 2x + 4)$

Solution:

$$\lim_{x \rightarrow 1} (3x^2 - 2x + 4)$$

$$\because \lim_{x \rightarrow a} [f(x) \pm g(x)] = \lim_{x \rightarrow a} f(x) \pm \lim_{x \rightarrow a} g(x)$$

$$= \lim_{x \rightarrow 1} (3x^2) - \lim_{x \rightarrow 1} (2x) + \lim_{x \rightarrow 1} (4)$$

$$\because \lim_{x \rightarrow a} [kf(x)] = k \left[\lim_{x \rightarrow a} f(x) \right]$$

$$= 3 \lim_{x \rightarrow 1} (x^2) - 2 \lim_{x \rightarrow 1} (x) + 4$$

$$= 3(1)^2 - 2(1) + 4$$

$$= 3 - 2 + 4$$

$$= 5$$

$$\boxed{\lim_{x \rightarrow 1} (3x^2 - 2x + 4) = 5}$$

(iii) $\lim_{x \rightarrow 3} \sqrt{x^2 + x + 4}$

Solution:

$$\lim_{x \rightarrow 3} \sqrt{x^2 + x + 4}$$

$$\because \lim_{x \rightarrow a} [f(x)]^n = \left[\lim_{x \rightarrow a} f(x) \right]^n$$

$$= \sqrt{\lim_{x \rightarrow 3} (x^2 + x + 4)}$$

$$\because \lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$$

$$= \sqrt{\lim_{x \rightarrow 3} x^2 + \lim_{x \rightarrow 3} x + \lim_{x \rightarrow 3} 4}$$

$$= \sqrt{3^2 + 3 + 4}$$

$$= \sqrt{9 + 3 + 4}$$

$$= \sqrt{16}$$

$$= 4$$

$$\boxed{\lim_{x \rightarrow 3} \sqrt{x^2 + x + 4} = 4}$$

(iv) $\lim_{x \rightarrow 2} x\sqrt{x^2 - 4}$

Solution:

$$\lim_{x \rightarrow 2} x\sqrt{x^2 - 4}$$

$$\because \lim_{x \rightarrow a} [f(x)g(x)] = \left[\lim_{x \rightarrow a} f(x) \right] \cdot \left[\lim_{x \rightarrow a} g(x) \right]$$

$$= \left[\lim_{x \rightarrow 2} x \right] \cdot \left[\lim_{x \rightarrow 2} \sqrt{x^2 - 4} \right]$$

$$\because \lim_{x \rightarrow a} [f(x)]^n = \left[\lim_{x \rightarrow a} f(x) \right]^n$$

$$= 2 \cdot \sqrt{\lim_{x \rightarrow 2} (x^2 - 4)}$$

$$\because \lim_{x \rightarrow a} [f(x) - g(x)] = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x)$$

$$= 2 \sqrt{\lim_{x \rightarrow 2} x^2 - \lim_{x \rightarrow 2} 4}$$

$$= 2 \sqrt{2^2 - 4}$$

$$= 2 \sqrt{4 - 4}$$

$$= 2 \sqrt{0}$$

$$= 2(0)$$

$$= 0$$

$$\boxed{\lim_{x \rightarrow 2} x\sqrt{x^2 - 4} = 0}$$

(v) $\lim_{x \rightarrow 2} (\sqrt{x^3 + 1} - \sqrt{x^2 + 5})$

Solution:

$$\lim_{x \rightarrow 2} (\sqrt{x^3 + 1} - \sqrt{x^2 + 5})$$

$$\because \lim_{x \rightarrow a} [f(x) - g(x)] = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x)$$

$$= \lim_{x \rightarrow 2} (\sqrt{x^3 + 1}) - \lim_{x \rightarrow 2} (\sqrt{x^2 + 5})$$

$$\begin{aligned}
 & \therefore \lim_{x \rightarrow a} [f(x)]^n = [\lim_{x \rightarrow a} f(x)]^n \\
 & = \sqrt{\lim_{x \rightarrow 2} (x^3 + 1)} - \sqrt{\lim_{x \rightarrow 2} (x^2 + 5)} \\
 & \therefore \lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x) \\
 & = \sqrt{\lim_{x \rightarrow 2} x^3 + \lim_{x \rightarrow 2} 1} - \sqrt{\lim_{x \rightarrow 2} x^2 + \lim_{x \rightarrow 2} 5} \\
 & = \sqrt{2^3 + 1} - \sqrt{2^2 + 5} \\
 & = \sqrt{9} - \sqrt{9} \\
 & \boxed{\lim_{x \rightarrow 2} (\sqrt{x^3 + 1} - \sqrt{x^2 + 5}) = 0}
 \end{aligned}$$

(vi) $\lim_{x \rightarrow -2} \frac{2x^3 + 5x}{3x - 2}$

Solution:

$$\begin{aligned}
 & \lim_{x \rightarrow -2} \frac{2x^3 + 5x}{3x - 2} \\
 & \therefore \lim_{x \rightarrow a} \left[\frac{f(x)}{g(x)} \right] = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} \\
 & = \frac{\lim_{x \rightarrow -2} (2x^3 + 5x)}{\lim_{x \rightarrow -2} (3x - 2)} \\
 & \therefore \lim_{x \rightarrow a} [f(x) \pm g(x)] = \lim_{x \rightarrow a} f(x) \pm \lim_{x \rightarrow a} g(x) \\
 & = \frac{\lim_{x \rightarrow -2} (2x^3) + \lim_{x \rightarrow -2} (5x)}{\lim_{x \rightarrow -2} (3x) - \lim_{x \rightarrow -2} (2)} \\
 & \therefore \lim_{x \rightarrow a} [kf(x)] = k \left[\lim_{x \rightarrow a} f(x) \right] \\
 & = \frac{2 \lim_{x \rightarrow -2} (x^3) + 5 \lim_{x \rightarrow -2} (x)}{3 \lim_{x \rightarrow -2} (x) - \lim_{x \rightarrow -2} (2)} \\
 & = \frac{2(-2)^3 + 5(-2)}{3(-2) - 2} \\
 & = \frac{2(-8) + 5(-2)}{3(-2) - 2} \\
 & = \frac{-16 - 10}{-6 - 2} \\
 & = \frac{-26}{-8} \\
 & = \frac{13}{4}
 \end{aligned}$$

Hence $\boxed{\lim_{x \rightarrow -2} \frac{2x^3 + 5x}{3x - 2} = \frac{13}{4}}$

Q.2 Evaluate each limit by using algebraic techniques.

(i) $\lim_{x \rightarrow -1} \frac{x^3 - x}{x + 1}$

Solution:

$$\begin{aligned}
 & \lim_{x \rightarrow -1} \frac{x^3 - x}{x + 1} \\
 & = \lim_{x \rightarrow -1} \frac{x(x^2 - 1)}{x + 1} \\
 & = \lim_{x \rightarrow -1} \frac{x(x - 1)(x + 1)}{x + 1} \\
 & = \lim_{x \rightarrow -1} x(x - 1) \\
 & = -1(-1 - 1) \\
 & = -1(-2) \\
 & = 2
 \end{aligned}$$

$$\boxed{\lim_{x \rightarrow -1} \frac{x^3 - x}{x + 1} = 2}$$

(ii) $\lim_{x \rightarrow 0} \left(\frac{3x^3 + 4x}{x^2 + x} \right)$

Solution:

$$\begin{aligned}
 & \lim_{x \rightarrow 0} \left(\frac{3x^3 + 4x}{x^2 + x} \right) \\
 & = \lim_{x \rightarrow 0} \frac{x(3x^2 + 4)}{x(x + 1)} \\
 & = \lim_{x \rightarrow 0} \frac{3x^2 + 4}{x + 1} \\
 & = \frac{3(0)^2 + 4}{0 + 1} \\
 & = \frac{0 + 4}{0 + 1} \\
 & = 4
 \end{aligned}$$

$$\boxed{\lim_{x \rightarrow 0} \frac{3x^3 + 4x}{x^2 + x} = 4}$$

(iii) $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 + x - 6}$

Solution:

$$\begin{aligned} & \lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 + x - 6} \\ &= \lim_{x \rightarrow 2} \frac{x^3 - 2^3}{x^2 + 3x - 2x - 6} \\ & \quad \because a^3 - b^3 = (a - b)(a^2 + ab + b^2) \\ &= \lim_{x \rightarrow 2} \frac{(x - 2)(x^2 + 2x + 2^2)}{x(x + 3) - 2(x + 3)} \\ &= \lim_{x \rightarrow 2} \frac{(x - 2)(x^2 + 2x + 4)}{(x + 3)(x - 2)} \\ &= \lim_{x \rightarrow 2} \frac{x^2 + 2x + 4}{x + 3} \\ &= \frac{2^2 + 2(2) + 4}{2 + 3} \\ &= \frac{4 + 4 + 4}{5} \\ &= \frac{12}{5} \end{aligned}$$

(iv) $\lim_{x \rightarrow 1} \frac{x^3 - 3x^2 + 3x - 1}{x^3 - x}$

Solution:

$$\begin{aligned} & \lim_{x \rightarrow 1} \frac{x^3 - 3x^2 + 3x - 1}{x^3 - x} \\ &= \lim_{x \rightarrow 1} \frac{(x^3) - 3(x)^2(1) + 3(x)(1)^2 - (1)^3}{x(x^2 - 1)} \\ & \quad \because (a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3 \\ &= \lim_{x \rightarrow 1} \frac{(x - 1)^3}{x(x^2 - 1)} \\ &= \lim_{x \rightarrow 1} \frac{(x - 1)^3}{x(x - 1)(x + 1)} \\ &= \lim_{x \rightarrow 1} \frac{(x - 1)^2}{x(x + 1)} \end{aligned}$$

$$= \frac{(1 - 1)^2}{1(1 + 1)}$$

$$= \frac{0}{2}$$

$$= 0$$

$$\boxed{\lim_{x \rightarrow 1} \frac{x^3 - 3x^2 + 3x - 1}{x^3 - x} = 0}$$

(v) $\lim_{x \rightarrow -1} \frac{x^3 + x^2}{x^2 - 1}$

Solution:

$$\begin{aligned} & \lim_{x \rightarrow -1} \frac{x^3 + x^2}{x^2 - 1} \\ &= \lim_{x \rightarrow -1} \frac{x^2(x + 1)}{(x - 1)(x + 1)} \\ &= \lim_{x \rightarrow -1} \frac{x^2}{x - 1} \\ &= \frac{(-1)^2}{-1 - 1} \\ &= -\frac{1}{2} \end{aligned}$$

$$\boxed{\lim_{x \rightarrow -1} \left(\frac{x^3 + x^2}{x^2 - 1} \right) = -\frac{1}{2}}$$

(vi) $\lim_{x \rightarrow 4} \frac{2x^2 - 32}{x^3 - 4x^2}$

Solution:

$$\begin{aligned} & \lim_{x \rightarrow 4} \frac{2x^2 - 32}{x^3 - 4x^2} \\ &= \lim_{x \rightarrow 4} \frac{2(x^2 - 16)}{x^2(x - 4)} \\ &= \lim_{x \rightarrow 4} \frac{2(x^2 - 4^2)}{x^2(x - 4)} \\ &= \lim_{x \rightarrow 4} \frac{2(x - 4)(x + 4)}{x^2(x - 4)} \\ &= \lim_{x \rightarrow 4} \frac{2(x + 4)}{x^2} \\ &= \frac{2(4 + 4)}{4^2} \end{aligned}$$

$$= \frac{2 \times 8}{16}$$

$$= 1$$

$$\boxed{\lim_{x \rightarrow 4} \left(\frac{2x^2 - 32}{x^3 - 4x^2} \right) = 1}$$

(vii) $\lim_{x \rightarrow 2} \frac{\sqrt{x} - \sqrt{2}}{x - 2}$

Solution:

$$\lim_{x \rightarrow 2} \frac{\sqrt{x} - \sqrt{2}}{x - 2}$$

By Rationalizing the numerator

$$= \lim_{x \rightarrow 2} \frac{\sqrt{x} - \sqrt{2}}{x - 2} \times \frac{\sqrt{x} + \sqrt{2}}{\sqrt{x} + \sqrt{2}}$$

$$= \lim_{x \rightarrow 2} \frac{(\sqrt{x})^2 - (\sqrt{2})^2}{(x - 2)(\sqrt{x} + \sqrt{2})}$$

$$= \lim_{x \rightarrow 2} \frac{x - 2}{(x - 2)(\sqrt{x} + \sqrt{2})}$$

$$= \lim_{x \rightarrow 2} \frac{1}{\sqrt{x} + \sqrt{2}}$$

$$= \frac{1}{\sqrt{2} + \sqrt{2}}$$

$$= \frac{1}{2\sqrt{2}}$$

$$\boxed{\lim_{x \rightarrow 2} \frac{\sqrt{x} - \sqrt{2}}{x - 2} = \frac{1}{2\sqrt{2}}}$$

(viii) $\lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h}$

Solution:

$$\lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h}$$

By rationalizing the numerator

$$= \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \times \frac{\sqrt{x+h} + \sqrt{x}}{\sqrt{x+h} + \sqrt{x}}$$

$$= \lim_{h \rightarrow 0} \frac{(\sqrt{x+h})^2 - (\sqrt{x})^2}{h[\sqrt{x+h} + \sqrt{x}]}$$

$$= \lim_{h \rightarrow 0} \frac{x + h - x}{h[\sqrt{x+h} + \sqrt{x}]}$$

$$= \lim_{h \rightarrow 0} \frac{h}{h[\sqrt{x+h} + \sqrt{x}]}$$

$$= \lim_{h \rightarrow 0} \frac{1}{\sqrt{x+h} + \sqrt{x}}$$

$$= \frac{1}{\sqrt{x+0} + \sqrt{x}}$$

$$= \frac{1}{\sqrt{x} + \sqrt{x}}$$

$$= \frac{1}{2\sqrt{x}}$$

$$\boxed{\lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} = \frac{1}{2\sqrt{x}}}$$

(ix) $\lim_{x \rightarrow a} \frac{x^n - a^n}{x^m - a^m}$

Solution:

$$\lim_{x \rightarrow a} \frac{x^n - a^n}{x^m - a^m}$$

$$= \lim_{x \rightarrow a} \frac{(x-a)(x^{n-1} + x^{n-2}a + x^{n-3}a^2 + \dots + a^{n-1})}{(x-a)(x^{m-1} + x^{m-2}a + x^{m-3}a^2 + \dots + a^{m-1})}$$

$$= \lim_{x \rightarrow a} \frac{x^{n-1} + x^{n-2}a + x^{n-3}a^2 + \dots + a^{n-1}}{x^{m-1} + x^{m-2}a + x^{m-3}a^2 + \dots + a^{m-1}}$$

$$= \frac{a^{n-1} + a^{n-2}a + a^{n-3}a^2 + \dots + a^{n-1}}{a^{m-1} + a^{m-2}a + a^{m-3}a^2 + \dots + a^{m-1}}$$

$$= \frac{a^{n-1} + a^{n-1} + a^{n-1} + \dots + a^{n-1}}{a^{m-1} + a^{m-1} + a^{m-1} + \dots + a^{m-1}}$$

$$= \frac{na^{n-1}}{ma^{m-1}}$$

$$= \frac{na^{n-1-(m-1)}}{m}$$

$$= \frac{na^{n-1-m+1}}{m}$$

$$= \frac{n}{m} a^{n-m}$$

$$\boxed{\lim_{x \rightarrow a} \frac{x^n - a^n}{x^m - a^m} = \frac{n}{m} a^{n-m}}$$

Q.3 Evaluate the following limits:

(i) $\lim_{x \rightarrow 0} \frac{\sin 7x}{x}$

Solution:

$$\lim_{x \rightarrow 0} \frac{\sin(7x)}{x}$$

Multiply and divide by 7

$$= \lim_{x \rightarrow 0} 7 \times \frac{\sin 7x}{7x}$$

$$= 7 \lim_{x \rightarrow 0} \frac{\sin 7x}{7x}$$

As $x \rightarrow 0$, then $7x \rightarrow 0$

$$= 7 \lim_{7x \rightarrow 0} \frac{\sin 7x}{7x}$$

$$= 7(1)$$

$$\boxed{\lim_{x \rightarrow 0} \frac{\sin 7x}{x} = 7}$$

(ii) $\lim_{x \rightarrow 0} \frac{\sin x^\circ}{x}$

Solution:

$$\lim_{x \rightarrow 0} \frac{\sin x^\circ}{x}$$

As $1^\circ = \frac{\pi}{180}$ radian

$$x^\circ = \frac{\pi x}{180} \text{ radian}$$

$$\lim_{x \rightarrow 0} \frac{\sin x^\circ}{x} = \lim_{x \rightarrow 0} \frac{\sin\left(\frac{\pi x}{180}\right)}{x}$$

Multiply and divide by $\frac{\pi}{180}$

$$= \lim_{x \rightarrow 0} \frac{\pi}{180} \times \frac{\sin\left(\frac{\pi x}{180}\right)}{\frac{\pi}{180} x}$$

$$= \frac{\pi}{180} \lim_{x \rightarrow 0} \frac{\sin\left(\frac{\pi x}{180}\right)}{\frac{\pi}{180} x}$$

When $x \rightarrow 0$, then $\frac{\pi}{180} x \rightarrow 0$

$$= \frac{\pi}{180} \lim_{\frac{\pi}{180} x \rightarrow 0} \frac{\sin\left(\frac{\pi}{180} x\right)}{\left(\frac{\pi}{180} x\right)}$$

$$= \frac{\pi}{180} \times 1$$

$$\boxed{\lim_{x \rightarrow 0} \frac{\sin x^\circ}{x} = \frac{\pi}{180}}$$

(iii) $\lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\sin \theta}$

Solution:

$$\lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\sin \theta}$$

By rationalizing the numerator

$$= \lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\sin \theta} \times \frac{1 + \cos \theta}{1 + \cos \theta}$$

$$= \lim_{\theta \rightarrow 0} \frac{1^2 - \cos^2 \theta}{\sin \theta (1 + \cos \theta)}$$

$$= \lim_{\theta \rightarrow 0} \frac{1 - \cos^2 \theta}{\sin \theta (1 + \cos \theta)}$$

$$= \lim_{\theta \rightarrow 0} \frac{\sin^2 \theta}{\sin \theta (1 + \cos \theta)}$$

$$= \lim_{\theta \rightarrow 0} \frac{\sin \theta}{1 + \cos \theta}$$

$$= \frac{\sin 0}{1 + \cos 0} = \frac{0}{1 + 1} = \frac{0}{2} = 0$$

$$\boxed{\lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\sin \theta} = 0}$$

(iv) $\lim_{x \rightarrow \pi} \frac{\sin x}{\pi - x}$

Solution:

$$\lim_{x \rightarrow \pi} \frac{\sin x}{\pi - x}$$

Let $\theta = \pi - x$

$$x = \pi - \theta$$

When $x \rightarrow \pi$, we have $\theta \rightarrow 0$ then

$$\lim_{x \rightarrow \pi} \frac{\sin x}{\pi - x} = \lim_{\theta \rightarrow 0} \frac{\sin(\pi - \theta)}{\theta}$$

$$\therefore \sin(\pi - \theta) = \sin \theta$$

$$= \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta}$$

$$= 1$$

$$\boxed{\lim_{x \rightarrow \pi} \frac{\sin x}{\pi - x} = 1}$$

(v) $\lim_{x \rightarrow 0} \frac{\sin ax}{\sin bx}$

Solution:

$$\lim_{x \rightarrow 0} \frac{\sin ax}{\sin bx}$$

$$= \lim_{x \rightarrow 0} \frac{\left(\frac{\sin ax}{ax} \right) \times ax}{\left(\frac{\sin bx}{bx} \right) \times bx}$$

$$= \frac{a}{b} \lim_{x \rightarrow 0} \frac{\left(\frac{\sin ax}{ax} \right)}{\left(\frac{\sin bx}{bx} \right)}$$

$$= \frac{a}{b} \frac{\lim_{x \rightarrow 0} \left(\frac{\sin ax}{ax} \right)}{\lim_{x \rightarrow 0} \left(\frac{\sin bx}{bx} \right)}$$

$$= \frac{a}{b} \left(\frac{1}{1} \right)$$

$$= \frac{a}{b}$$

$$\boxed{\lim_{x \rightarrow 0} \frac{\sin ax}{\sin bx} = \frac{a}{b}}$$

(vi) $\lim_{x \rightarrow 0} \frac{x}{\tan x}$

Solution:

$$\lim_{x \rightarrow 0} \frac{x}{\tan x}$$

$$= \lim_{x \rightarrow 0} \frac{x}{\left(\frac{\sin x}{\cos x} \right)}$$

$$= \lim_{x \rightarrow 0} \frac{\cos x}{\left(\frac{\sin x}{x} \right)}$$

$$\lim_{x \rightarrow 0} \cos x$$

$$= \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)$$

$$= \frac{\cos 0}{1}$$

$$= 1$$

$$\boxed{\lim_{x \rightarrow 0} \frac{x}{\tan x} = 1}$$

(vii) $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2}$

Solution:

$$\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2}$$

$$\because \cos 2x = 1 - 2\sin^2 x$$

$$= \lim_{x \rightarrow 0} \frac{2\sin^2 x}{x^2}$$

$$= 2 \lim_{x \rightarrow 0} \frac{\sin^2 x}{x^2}$$

$$= 2 \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^2$$

$$= 2 \left(\lim_{x \rightarrow 0} \frac{\sin x}{x} \right)^2$$

$$= 2(1)^2$$

$$= 2$$

$$\boxed{\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2} = 2}$$

(viii) $\lim_{x \rightarrow 0} \frac{1 - \cos x}{\sin^2 x}$

Solution:

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{\sin^2 x}$$

$$= \lim_{x \rightarrow 0} \frac{1 - \cos x}{1 - \cos^2 x}$$

$$= \lim_{x \rightarrow 0} \frac{1 - \cos x}{(1 - \cos x)(1 + \cos x)}$$

$$= \lim_{x \rightarrow 0} \frac{1}{1 + \cos x}$$

$$= \frac{1}{1 + \cos 0}$$

$$= \frac{1}{1+1}$$

$$= \frac{1}{2}$$

$$\boxed{\lim_{x \rightarrow 0} \frac{1 - \cos x}{\sin^2 x} = \frac{1}{2}}$$

$$(ix) \quad \lim_{\theta \rightarrow 0} \frac{\sin^2 \theta}{\theta}$$

Solution:

$$\lim_{\theta \rightarrow 0} \frac{\sin^2 \theta}{\theta}$$

$$= \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} \cdot \sin \theta$$

$$= \left(\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} \right) \left(\lim_{\theta \rightarrow 0} \sin \theta \right)$$

$$= 1 \times \sin 0$$

$$= 1 \times (0)$$

$$= 0$$

$$\boxed{\lim_{\theta \rightarrow 0} \frac{\sin^2 \theta}{\theta} = 0}$$

$$(x) \quad \lim_{x \rightarrow 0} \frac{\sec x - \cos x}{x}$$

Solution:

$$\lim_{x \rightarrow 0} \frac{\sec x - \cos x}{x}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{1}{\cos x} - \cos x}{x}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{1 - \cos^2 x}{\cos x}}{x}$$

$$= \lim_{x \rightarrow 0} \frac{\sin^2 x}{x \cos x}$$

$$= \lim_{x \rightarrow 0} \frac{\sin x}{x} \cdot \frac{\sin x}{\cos x}$$

$$= \left(\lim_{x \rightarrow 0} \frac{\sin x}{x} \right) \left(\lim_{x \rightarrow 0} \tan x \right)$$

$$= 1 \times \tan(0)$$

$$= 1 \times 0$$

$$= 0$$

$$\boxed{\lim_{x \rightarrow 0} \frac{\sec x - \cos x}{x} = 0}$$

$$(xi) \quad \lim_{\theta \rightarrow 0} \frac{1 - \cos p\theta}{1 - \cos q\theta}$$

Solution:

$$\lim_{\theta \rightarrow 0} \frac{1 - \cos p\theta}{1 - \cos q\theta}$$

$$\because 1 - \cos \theta = 2 \sin^2 \left(\frac{\theta}{2} \right)$$

$$= \lim_{\theta \rightarrow 0} \frac{2 \sin^2 \left(\frac{p\theta}{2} \right)}{2 \sin^2 \left(\frac{q\theta}{2} \right)}$$

$$= \lim_{\theta \rightarrow 0} \left[\frac{\sin \left(\frac{p\theta}{2} \right)}{\sin \left(\frac{q\theta}{2} \right)} \right]^2$$

$$= \lim_{\theta \rightarrow 0} \left[\frac{\frac{\sin \left(\frac{p\theta}{2} \right) \times \frac{p\theta}{2}}{\frac{p\theta}{2}}}{\frac{\sin \left(\frac{q\theta}{2} \right) \times \frac{q\theta}{2}}{\frac{q\theta}{2}}} \right]^2$$

$$= \lim_{\theta \rightarrow 0} \left[\frac{\frac{\sin \left(\frac{p\theta}{2} \right)}{\frac{p\theta}{2}} \times p}{\frac{\sin \left(\frac{q\theta}{2} \right)}{\frac{q\theta}{2}} \times q} \right]^2$$

$$= \frac{p^2}{q^2} \left[\frac{\lim_{\theta \rightarrow 0} \frac{\sin \left(\frac{p\theta}{2} \right)}{\frac{p\theta}{2}}}{\lim_{\theta \rightarrow 0} \frac{\sin \left(\frac{q\theta}{2} \right)}{\frac{q\theta}{2}}} \right]^2$$

$$= \frac{p^2}{q^2} \left(\frac{1}{1} \right)^2$$

$$= \frac{p^2}{q^2}$$

$$\boxed{\lim_{\theta \rightarrow 0} \frac{1 - \cos p\theta}{1 - \cos q\theta} = \frac{p^2}{q^2}}$$

(xii) $\lim_{\theta \rightarrow 0} \frac{\tan \theta - \sin \theta}{\sin^3 \theta}$

Solution:

$$\lim_{\theta \rightarrow 0} \frac{\tan \theta - \sin \theta}{\sin^3 \theta}$$

$$= \lim_{\theta \rightarrow 0} \frac{\frac{\sin \theta}{\cos \theta} - \sin \theta}{\sin^3 \theta}$$

$$= \lim_{\theta \rightarrow 0} \frac{\sin \theta \left(\frac{1}{\cos \theta} - 1 \right)}{\sin^3 \theta}$$

$$= \lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\cos \theta \sin^2 \theta}$$

$$= \lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\cos \theta (1 - \cos^2 \theta)}$$

$$= \lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\cos \theta (1 - \cos \theta)(1 + \cos \theta)}$$

$$= \lim_{\theta \rightarrow 0} \frac{1}{\cos \theta (1 + \cos \theta)}$$

$$= \frac{1}{\cos 0 (1 + \cos 0)}$$

$$= \frac{1}{1(1+1)}$$

$$= \frac{1}{2}$$

$$\boxed{\lim_{\theta \rightarrow 0} \frac{\tan \theta - \sin \theta}{\sin^3 \theta} = \frac{1}{2}}$$

Q.4 Express each limit in terms of e :

(i) $\lim_{n \rightarrow +\infty} \left(1 + \frac{1}{n} \right)^{2n}$

Solution:

$$\lim_{n \rightarrow +\infty} \left(1 + \frac{1}{n} \right)^{2n}$$

$$= \lim_{n \rightarrow +\infty} \left[\left(1 + \frac{1}{n} \right)^n \right]^2$$

$$= \left[\lim_{n \rightarrow +\infty} \left(1 + \frac{1}{n} \right)^n \right]^2$$

$$= e^2$$

$$\boxed{\lim_{n \rightarrow +\infty} \left(1 + \frac{1}{n} \right)^{2n} = e^2}$$

(ii) $\lim_{n \rightarrow +\infty} \left(1 + \frac{1}{n} \right)^{\frac{n}{2}}$

Solution:

$$\lim_{n \rightarrow +\infty} \left(1 + \frac{1}{n} \right)^{\frac{n}{2}}$$

$$= \lim_{n \rightarrow +\infty} \left[\left(1 + \frac{1}{n} \right)^n \right]^{\frac{1}{2}}$$

$$= \left[\lim_{n \rightarrow +\infty} \left(1 + \frac{1}{n} \right)^n \right]^{\frac{1}{2}}$$

$$= e^{\frac{1}{2}}$$

$$\boxed{\lim_{n \rightarrow +\infty} \left(1 + \frac{1}{n} \right)^{2n} = \sqrt{e}}$$

(iii) $\lim_{n \rightarrow +\infty} \left(1 - \frac{1}{n} \right)^n$

Solution:

$$\lim_{n \rightarrow +\infty} \left(1 - \frac{1}{n} \right)^n$$

$$= \lim_{n \rightarrow +\infty} \left[\left(1 - \frac{1}{n} \right)^{-n} \right]^{-1}$$

$$= \left[\lim_{n \rightarrow \infty} \left(1 + \left(-\frac{1}{n} \right) \right)^{-n} \right]^{-1}$$

$$= e^{-1}$$

$$= \frac{1}{e}$$

$$\boxed{\lim_{n \rightarrow +\infty} \left(1 - \frac{1}{n} \right)^n = \frac{1}{e}}$$

(iv) $\lim_{n \rightarrow +\infty} \left(1 + \frac{1}{3n} \right)^n$

Solution:

$$\lim_{n \rightarrow +\infty} \left(1 + \frac{1}{3n} \right)^n$$

$$= \lim_{n \rightarrow +\infty} \left[\left(1 + \frac{1}{3n} \right)^n \right]^{\frac{3}{3}}$$

$$= \left[\lim_{n \rightarrow \infty} \left(1 + \frac{1}{3n} \right)^{3n} \right]^{\frac{1}{3}}$$

$$\boxed{\lim_{n \rightarrow +\infty} \left(1 + \frac{1}{3n} \right)^n = e^{\frac{1}{3}}}$$

(v) $\lim_{n \rightarrow +\infty} \left(1 + \frac{4}{n} \right)^n$

Solution:

$$\lim_{n \rightarrow +\infty} \left(1 + \frac{4}{n} \right)^n$$

$$= \lim_{n \rightarrow \infty} \left[\left(1 + \frac{4}{n} \right)^n \right]^{\frac{4}{4}}$$

$$= \left(\lim_{n \rightarrow \infty} \left(1 + \frac{4}{n} \right)^{\frac{n}{4}} \right)^4$$

$$\boxed{\lim_{n \rightarrow +\infty} \left(1 + \frac{4}{n} \right)^n = e^4}$$

(vi) $\lim_{x \rightarrow 0} (1 + 3x)^{\frac{2}{x}}$

Solution:

$$\lim_{x \rightarrow 0} (1 + 3x)^{\frac{2}{x}}$$

$$= \lim_{x \rightarrow 0} \left[(1 + 3x)^{\frac{2}{x}} \right]^{\frac{3}{3}}$$

$$= \left[\lim_{x \rightarrow 0} (1 + 3x)^{\frac{1}{3x}} \right]^6$$

$$\boxed{\lim_{x \rightarrow 0} (1 + 3x)^{\frac{2}{x}} = e^6}$$

(vii) $\lim_{x \rightarrow 0} (1 + 2x^2)^{\frac{1}{x^2}}$

Solution:

$$\lim_{x \rightarrow 0} (1 + 2x^2)^{\frac{1}{x^2}}$$

$$= \lim_{x \rightarrow 0} \left[(1 + 2x^2)^{\frac{1}{x^2}} \right]^{\frac{2}{2}}$$

$$\boxed{\lim_{x \rightarrow 0} (1 + 2x^2)^{\frac{1}{x^2}} = e^2}$$

(viii) $\lim_{h \rightarrow 0} (1 - 2h)^{\frac{1}{h}}$

Solution:

$$\lim_{h \rightarrow 0} (1 - 2h)^{\frac{1}{h}}$$

$$= \lim_{h \rightarrow 0} \left[(1 - 2h)^{\frac{1}{h}} \right]^{\frac{-2}{-2}}$$

$$= \lim_{h \rightarrow 0} \left[(1 - 2h)^{-\frac{1}{2h}} \right]^{-2}$$

$$= \left[\lim_{h \rightarrow 0} (1 - 2h)^{-\frac{1}{2h}} \right]^{-2}$$

$$= e^{-2}$$

$$\boxed{\lim_{h \rightarrow 0} (1 - 2h)^{\frac{1}{h}} = \frac{1}{e^2}}$$

(ix) $\lim_{x \rightarrow \infty} \left(\frac{x}{1+x} \right)^x$

Solution:

$$\begin{aligned} & \lim_{x \rightarrow \infty} \left(\frac{x}{1+x} \right)^x \\ &= \lim_{x \rightarrow \infty} \left(\frac{1+x}{x} \right)^{-x} \\ &= \lim_{x \rightarrow \infty} \left(\frac{1}{x} + 1 \right)^{-x} \\ &= \lim_{x \rightarrow \infty} \left[\left(1 + \frac{1}{x} \right)^x \right]^{-1} \\ &= \left[\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x} \right)^x \right]^{-1} \\ &= e^{-1} \\ &= \frac{1}{e} \end{aligned}$$

$$\boxed{\lim_{x \rightarrow \infty} \left(\frac{x}{1+x} \right)^x = \frac{1}{e}}$$

(x) $\lim_{x \rightarrow 0} \frac{e^{\frac{1}{x}} - 1}{\frac{1}{x} + 1}, x < 0$

Solution:

$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{e^{\frac{1}{x}} - 1}{\frac{1}{x} + 1}, x < 0 \\ & \text{Replace } x \text{ by } -y \text{ where } y > 0 \\ &= \lim_{y \rightarrow 0} \frac{e^{-\frac{1}{y}} - 1}{-\frac{1}{y} + 1} \\ &= \lim_{y \rightarrow 0} \frac{e^{-\frac{1}{y}} - 1}{-\frac{1}{y} + 1} \\ &= \lim_{y \rightarrow 0} \frac{e^{-\frac{1}{y}} - 1}{-\frac{1}{y} + 1} \\ &= \lim_{y \rightarrow 0} \frac{e^{-\frac{1}{y}} - 1}{-\frac{1}{y} + 1} \\ &= \frac{e^0 - 1}{-\frac{1}{0} + 1} \\ &= \frac{1}{e^0} \end{aligned}$$

$$\begin{aligned} & \frac{1}{e^0} - 1 \\ &= \frac{1}{e^0} + 1 \end{aligned}$$

$$\begin{aligned} & \frac{1}{e^0} - 1 \\ &= \frac{1}{e^0} + 1 \end{aligned}$$

$$\begin{aligned} & \frac{0-1}{0+1} \end{aligned}$$

$$\boxed{\lim_{x \rightarrow 0} \frac{e^{\frac{1}{x}} - 1}{\frac{1}{x} + 1}, x < 0 = -1}$$

(xi) $\lim_{x \rightarrow 0} \frac{e^{\frac{1}{x}} - 1}{\frac{1}{x} + 1}, x > 0$

Solution:

$$\lim_{x \rightarrow 0} \frac{e^{\frac{1}{x}} - 1}{\frac{1}{x} + 1}$$

As $x > 0$, so $x = x$

$$\begin{aligned} & e^{\frac{1}{x}} \left[1 - \frac{1}{\frac{1}{x}} \right] \\ &= \lim_{x \rightarrow 0} \frac{e^{\frac{1}{x}} \left[1 - \frac{1}{\frac{1}{x}} \right]}{e^{\frac{1}{x}} \left[1 + \frac{1}{\frac{1}{x}} \right]} \end{aligned}$$

$$\begin{aligned} & 1 - \frac{1}{\frac{1}{x}} \\ &= \lim_{x \rightarrow 0} \frac{e^{\frac{1}{x}}}{1 + \frac{1}{\frac{1}{x}}} \end{aligned}$$

$$\begin{aligned} & 1 - \frac{1}{\frac{1}{0}} \\ &= \frac{e^0}{1 + \frac{1}{\frac{1}{0}}} \end{aligned}$$

$$\begin{aligned}
 &= \frac{1 - \frac{1}{e^\infty}}{1 + \frac{1}{e^\infty}} \\
 &= \frac{1 - \frac{1}{\infty}}{1 + \frac{1}{\infty}} \\
 &= \frac{1 - 0}{1 + 0}
 \end{aligned}$$

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{e^x + 1}, x > 0 = 1$$

Left Hand Limit:

$\lim_{x \rightarrow c^-} f(x) = L$ is read as the limit of $f(x)$ is equal to L as x approaches c from the left i.e., For all x sufficiently close to c , but less than c , the value of $f(x)$ can be made as close as we please to L .

Right Hand Limit:

$\lim_{x \rightarrow c^+} f(x) = M$ is read as the limit of $f(x)$ is equal to M as x approaches from the right i.e., for all x sufficiently close to c , but greater than c , the value of $f(x)$ can be made as close as we please to M .

Criterion for Existence of Limit of a Function:

$$\lim_{x \rightarrow c} f(x) = L \text{ iff } \lim_{x \rightarrow c^-} f(x) = \lim_{x \rightarrow c^+} f(x) = L$$

Continuous Function:

A function f is said to be **continuous** at a number " c " iff the following three conditions are satisfied:

- (i) $f(c)$ is defined.
- (ii) $\lim_{x \rightarrow c} f(x)$ exists.
- (iii) $\lim_{x \rightarrow c} f(x) = f(c)$

Discontinuous Function:

If one or more of these three conditions fail to hold at c then the function f is said to be **discontinuous** at c .

Example: Discuss the continuity of the function $f(x)$ and $g(x)$ at $x = 3$.

$$\begin{aligned}
 \text{(a)} \quad f(x) &= \begin{cases} \frac{x^2 - 9}{x - 3} & \text{if } x \neq 3 \\ 6 & \text{if } x = 3 \end{cases} & \text{(b)} \quad g(x) &= \frac{x^2 - 9}{x - 3} \text{ if } x \neq 3.
 \end{aligned}$$

Solution:

$$\text{(a)} \quad \text{Given } f(3) = 6$$

∴ The function f is defined at $x = 3$.

$$\begin{aligned} \text{Now } \lim_{x \rightarrow 3} f(x) &= \lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} \\ &= \lim_{x \rightarrow 3} \frac{(x+3)(x-3)}{x-3} \\ &= \lim_{x \rightarrow 3} (x+3) = 6 \end{aligned}$$

$$\text{As } \lim_{x \rightarrow 3} f(x) = 6 = f(3)$$

∴ $f(x)$ is continuous at $x = 3$

It is noted that there is no break in the graph.

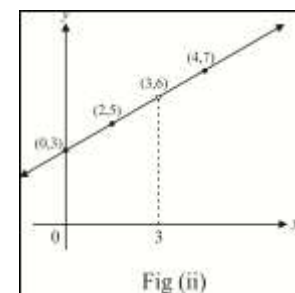
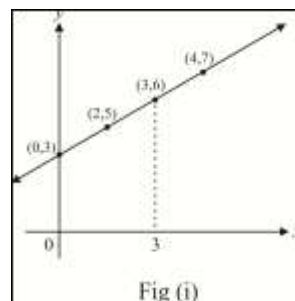
(See figure (i))

$$(b) \quad g(x) = \frac{x^2 - 9}{x - 3} \text{ if } x \neq 3$$

As $g(x)$ is not defined at $x = 3$

$g(x)$ is discontinuous at $x = 3$

It is noted that there is a break in the graph at $x = 3$. (See figure (ii))



EXERCISE 1.4

Q.1 Determine the left hand limit and the right hand limit and then, find the limit of the following functions when $x \rightarrow c$.

(i) $f(x) = 2x^2 + x - 5, c = 1$

Solution:

$$f(x) = 2x^2 + x - 5$$

Left hand limit:

$$\begin{aligned} \lim_{x \rightarrow 1^-} f(x) &= \lim_{x \rightarrow 1^-} (2x^2 + x - 5) \\ &= 2(1)^2 + 1 - 5 \\ &= 2 + 1 - 5 \\ &= -2 \end{aligned}$$

Right hand limit:

$$\begin{aligned} \lim_{x \rightarrow 1^+} f(x) &= \lim_{x \rightarrow 1^+} (2x^2 + x - 5) \\ &= 2(1)^2 + 1 - 5 \\ &= 2 + 1 - 5 \\ &= -2 \end{aligned}$$

$$\text{As } \lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x)$$

Hence $\lim_{x \rightarrow 1} f(x)$ exists and

$$\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} (2x^2 + x - 5) = -2$$

(ii) $f(x) = \frac{x^2 - 9}{x - 3}, c = -3$

Solution:

$$f(x) = \frac{x^2 - 9}{x - 3}, c = -3$$

Left hand limit:

$$\begin{aligned} \lim_{x \rightarrow -3^-} f(x) &= \lim_{x \rightarrow -3^-} \frac{x^2 - 9}{x - 3} \\ &= \frac{(-3)^2 - 9}{-3 - 3} \\ &= \frac{9 - 9}{-6} \\ &= \frac{0}{-6} \\ &= 0 \end{aligned}$$

Right hand limit:

$$\lim_{x \rightarrow -3^+} f(x) = \lim_{x \rightarrow -3^+} \frac{x^2 - 9}{x - 3}$$

$$\begin{aligned}
 &= \frac{(-3)^2 - 9}{-3 - 3} \\
 &= \frac{9 - 9}{-6} \\
 &= \frac{0}{-6} \\
 &= 0
 \end{aligned}$$

As $\lim_{x \rightarrow -3^-} f(x) = \lim_{x \rightarrow -3^+} f(x)$

Hence $\lim_{x \rightarrow c} f(x)$ exists

and $\lim_{x \rightarrow -3} f(x) = \lim_{x \rightarrow -3} \frac{x^2 - 9}{x - 3} = 0$

(iii) $f(x) = |x - 5|, \quad c = 5$

Solution:

$f(x) = |x - 5|, \quad c = 5$

Left hand limit:

$$\begin{aligned}
 \lim_{x \rightarrow 5^-} f(x) &= \lim_{x \rightarrow 5^-} |x - 5| \\
 &= \lim_{x \rightarrow 5^-} [-(x - 5)] \\
 &= -(5 - 5) \\
 &= 0
 \end{aligned}$$

Right hand limit:

$$\begin{aligned}
 \lim_{x \rightarrow 5^+} f(x) &= \lim_{x \rightarrow 5^+} |x - 5| \\
 &= \lim_{x \rightarrow 5^+} (x - 5) \\
 &= 5 - 5 \\
 &= 0
 \end{aligned}$$

As $\lim_{x \rightarrow 5^-} f(x) = \lim_{x \rightarrow 5^+} f(x)$

Hence $\lim_{x \rightarrow 5} f(x)$ exists and

$$\lim_{x \rightarrow 5} f(x) = \lim_{x \rightarrow 5} |x - 5| = 0$$

Q.2 Discuss the continuity of $f(x)$ at $x = c$:

(i) $f(x) = \begin{cases} 2x + 5 & \text{if } x \leq 2 \\ 4x + 1 & \text{if } x > 2 \end{cases}, \quad c = 2$

Solution:

At $x = 2$

$$f(x) = 2x + 5$$

$$f(2) = 2(2) + 5 = 9$$

$$f(2) = 9$$

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} (2x + 5)$$

$$= 2(2) + 5$$

$$= 9$$

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} (4x + 1)$$

$$= 4(2) + 1$$

$$= 9$$

As $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x)$, so

$\lim_{x \rightarrow 2} f(x)$ exists.

As $f(2) = \lim_{x \rightarrow 2} f(x)$

Hence, $f(x)$ is continuous at $x = 2$.

(ii) $f(x) = \begin{cases} 3x - 1 & \text{if } x < 1 \\ 4 & \text{if } x = 1, \quad c = 1 \\ 2x & \text{if } x > 1 \end{cases}$

Solution:

At $x = 1$

$$f(x) = 4$$

$$f(1) = 4$$

$$\begin{aligned}
 \lim_{x \rightarrow 1^-} f(x) &= \lim_{x \rightarrow 1^-} (3x - 1) \\
 &= 3(1) - 1 = 2
 \end{aligned}$$

$$\begin{aligned}
 \lim_{x \rightarrow 1^+} f(x) &= \lim_{x \rightarrow 1^+} (2x) \\
 &= 2 \lim_{x \rightarrow 1^+} (x) = 2
 \end{aligned}$$

As $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x)$,

so $\lim_{x \rightarrow 2} f(x)$ exists.

As, $f(1) \neq \lim_{x \rightarrow 1} f(x)$

$$4 \neq 2$$

Hence function $f(x)$ is discontinuous at $x = 1$

Q.3 If $f(x) = \begin{cases} 3x & \text{if } x \leq -2 \\ x^2 - 1 & \text{if } -2 < x < 2 \\ 3 & \text{if } x \geq 2 \end{cases}$

Discuss continuity at $x = 2$ and $x = -2$.

Solution:

Continuity at $x = 2$

At $x = 2$

$$f(2) = 3$$

$$\begin{aligned}
 \lim_{x \rightarrow 2^-} f(x) &= \lim_{x \rightarrow 2^-} (x^2 - 1) \\
 &= 2^2 - 1 = 4 - 1
 \end{aligned}$$

$$= 3$$

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} (3)$$

$$= 3$$

As $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x)$, so

$\lim_{x \rightarrow 2} f(x)$ exists.

$$\text{As } f(2) = \lim_{x \rightarrow 2} f(x)$$

Hence, $f(x)$ is continuous at $x = 2$.

continuity at $x = -2$

at $x = -2$

$$f(x) = 3x$$

$$f(-2) = 3(-2) = -6$$

$$\lim_{x \rightarrow -2^-} f(x) = \lim_{x \rightarrow -2^-} (3x)$$

$$= 3(-2)$$

$$= -6$$

$$\lim_{x \rightarrow -2^+} f(x) = \lim_{x \rightarrow -2^+} (x^2 - 1)$$

$$= (-2)^2 - 1$$

$$= 4 - 1$$

$$= 3$$

As $\lim_{x \rightarrow 2^-} f(x) \neq \lim_{x \rightarrow 2^+} f(x)$,

so $\lim_{x \rightarrow 2} f(x)$ does not exist.

Hence, $f(x)$ is discontinuous at $x = -2$.

Q.4 If $f(x) = \begin{cases} x+2 & , \quad x \leq -1 \\ c+2 & , \quad x > -1 \end{cases}$

Find “c” so that $\lim_{x \rightarrow -1} f(x)$ exists.

Solution:

As $\lim_{x \rightarrow -1} f(x)$ exists.

$$\text{So } \lim_{x \rightarrow -1^-} f(x) = \lim_{x \rightarrow -1^+} f(x)$$

$$\lim_{x \rightarrow -1^-} (x+2) = \lim_{x \rightarrow -1^+} (c+2)$$

$$-1+2 = c+2$$

$$-1 = c$$

$$\boxed{c = -1}$$

Q.5 Find the values m and n , so that given function f is continuous at $x = 3$.

$$(i) \quad f(x) = \begin{cases} mx & \text{if } x < 3 \\ n & \text{if } x = 3 \\ -2x+9 & \text{if } x > 3 \end{cases}$$

Solution:

As $f(x)$ is continuous at $x = 3$

$$\text{So } f(3) = \lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^+} f(x)$$

At $x = 3$

$$f(3) = n$$

$$\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^-} (mx)$$

$$= m(3)$$

$$= 3m$$

$$\lim_{x \rightarrow 3^+} f(x) = \lim_{x \rightarrow 3^+} (-2x+9)$$

$$= -2(3)+9$$

$$= -6+9$$

$$= 3$$

$$\text{As } f(3) = \lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^+} f(x)$$

$$n = 3m = 3$$

$$\boxed{n = 3}, \quad 3m = 3$$

$$\boxed{m = 1}$$

$$(ii) \quad f(x) = \begin{cases} mx & \text{if } x < 3 \\ x^2 & \text{if } x \geq 3 \end{cases}$$

Solution:

As $f(x)$ is continuous at $x = 3$

$$\text{So } f(3) = \lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^+} f(x)$$

At $x = 3$

$$f(x) = x^2$$

$$f(3) = 3^2 = 9$$

$$\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^-} (mx)$$

$$= 3m$$

$$\lim_{x \rightarrow 3^+} f(x) = \lim_{x \rightarrow 3^+} (x^2)$$

$$= 3^2$$

$$= 9$$

$$\text{As } f(3) = \lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^+} f(x)$$

$$9 = 3m = 9$$

$$3m = 3$$

$$\boxed{m=1}$$

Q.6 If

$$f(x) = \begin{cases} \frac{\sqrt{2x+5} - \sqrt{x+7}}{x-2}, & x \neq 2 \\ k, & x = 2 \end{cases}$$

find the value of k so that f is continuous at $x = 2$.

Solution:

As $f(x)$ is continuous at $x = 2$

$$\text{So } f(2) = \lim_{x \rightarrow 2} f(x)$$

At $x = 2$

$$f(2) = k$$

Now,

$$\lim_{x \rightarrow 2} f(x)$$

$$= \lim_{x \rightarrow 2} \frac{\sqrt{2x+5} - \sqrt{x+7}}{x-2}$$

By rationalization of numerator

$$= \lim_{x \rightarrow 2} \frac{\sqrt{2x+5} - \sqrt{x+7}}{x-2} \times \frac{\sqrt{2x+5} + \sqrt{x+7}}{\sqrt{2x+5} + \sqrt{x+7}}$$

$$= \lim_{x \rightarrow 2} \frac{(\sqrt{2x+5})^2 - (\sqrt{x+7})^2}{(x-2)(\sqrt{2x+5} + \sqrt{x+7})}$$

$$= \lim_{x \rightarrow 2} \frac{2x+5 - (x+7)}{(x-2)(\sqrt{2x+5} + \sqrt{x+7})}$$

$$= \lim_{x \rightarrow 2} \frac{2x+5 - x - 7}{(x-2)(\sqrt{2x+5} + \sqrt{x+7})}$$

$$= \lim_{x \rightarrow 2} \frac{x-2}{(x-2)(\sqrt{2x+5} + \sqrt{x+7})}$$

$$= \lim_{x \rightarrow 2} \frac{1}{\sqrt{2x+5} + \sqrt{x+7}}$$

$$= \frac{1}{\sqrt{2(2)+5} + \sqrt{2+7}}$$

$$= \frac{1}{\sqrt{9} + \sqrt{9}}$$

$$= \frac{1}{3+3}$$

$$\lim_{x \rightarrow 2} f(x) = \frac{1}{6}$$

$$\text{As } f(2) = \lim_{x \rightarrow 2} f(x)$$

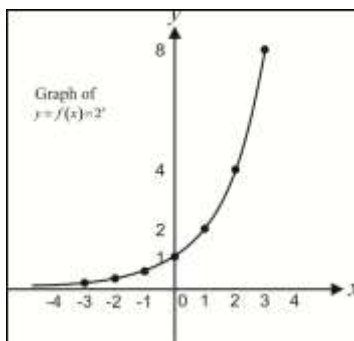
$$\boxed{k = \frac{1}{6}}$$

Graph of the Exponential Function $f(x) = a^x$:

Let us draw the graph of $y = 2^x$, here $a = 2$.

We prepare the following table for different values of x and $f(x)$ near the origin:

x	-4	-3	-2	-1	0	1	2	3	4
y	0.0625	0.125	0.25	0.5	1	2	4	8	16



Plotting the points (x, y) and joining them with smooth curve as shown in the figure, we get the graph of $y = 2^x$

From the graph of 2^x , the characteristics of the graph of $y = a^x$ are observed as follows:

If $a > 1$,

- (i) a^x is always positive for all real values of x .
- (ii) a^x increases as x increases.
- (iii) $a^x = 1$ when $x = 0$
- (iv) $a^x \rightarrow 0$ as $x \rightarrow -\infty$

Graph of Common Logarithmic Function $f(x) = \log x$:

If $x = 10^y$, then $y = \log x$

Now for all real values of $y, 10^y > 0 \Rightarrow x > 0$

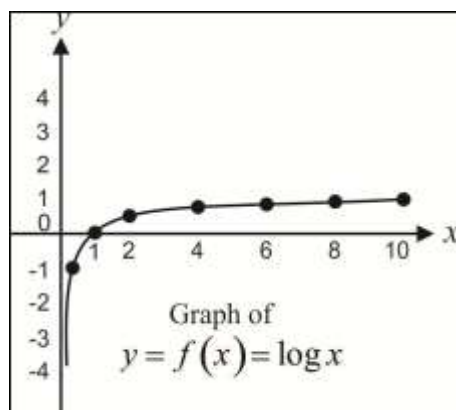
This means $\log x$ exists only when $x > 0$

$\Rightarrow$ Domain of the $\log x$ is positive real numbers. It is undefined at $x = 0$.

For graph of $f(x) = \log x$, we find the values of $\lg x$ from the common logarithmic table for various values of $x > 0$.

Table of some of the corresponding values of x and $f(x)$ is as under.

x	$\rightarrow 0$	0.1	1	2	4	6	8	10	$\rightarrow +\infty$
$y = f(x) = \log x$	$\rightarrow -\infty$	-1	0	0.30	0.60	0.77	0.90	1	$\rightarrow +\infty$



Plotting the points (x, y) and joining them with a smooth curve we get the graph as shown in the figure.

Note:

- (i) If we replace (x, y) with $(x, -y)$ and there is no change in the equation then the graph is symmetric with respect to x -axis.
- (ii) If we replace (x, y) with $(-x, y)$ and there is no change in the equation then the graph is symmetric with respect to y -axis.
- (iii) If we replace (x, y) with $(-x, -y)$ and there is no change in the equation then the graph is symmetric with respect to origin.

EXERCISE 1.5**Q.1 Draw the graphs of the following equations**

(i) $x^2 + y^2 = 9$ (ii) $\frac{x^2}{16} + \frac{y^2}{4} = 1$ (iii) $y = e^{2x}$ (iv) $y = 3^x$

(i) $x^2 + y^2 = 9$

Solution:

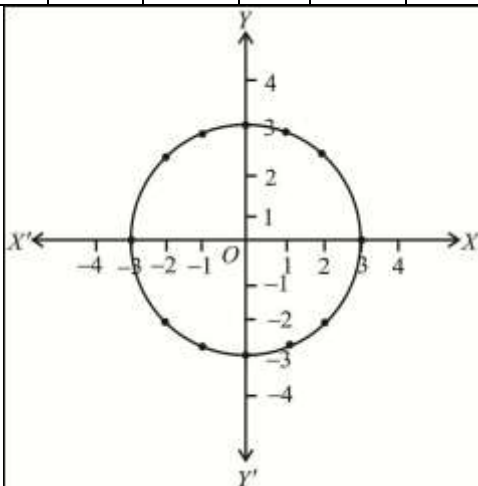
$$x^2 + y^2 = 9$$

$$y^2 = 9 - x^2$$

$$y = \pm\sqrt{9 - x^2}$$

Here domain = $[-3, 3]$

X	-3	-2	-1	0	1	2	3
Y	0	± 2.2	± 2.8	± 3	± 2.8	± 2.2	0



(ii) $\frac{x^2}{16} + \frac{y^2}{4} = 1$

Solution:

Given $\frac{x^2}{16} + \frac{y^2}{4} = 1$

$$\frac{y^2}{4} = 1 - \frac{x^2}{16}$$

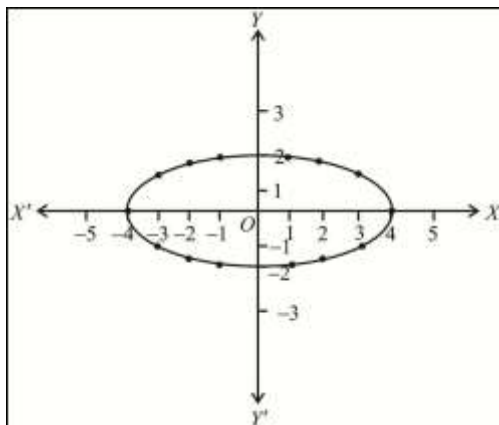
$$\frac{y^2}{4} = \frac{16 - x^2}{16}$$

$$y^2 = 4 \left(\frac{16 - x^2}{16} \right) = \frac{16 - x^2}{4}$$

$$y = \pm \frac{\sqrt{16 - x^2}}{2}$$

Here domain = $[-4, 4]$

x	-4	-3	-2	-1	0	1	2	3	4
y	0	± 1.3	± 1.7	± 1.9	± 2	± 1.9	± 1.7	± 1.3	0

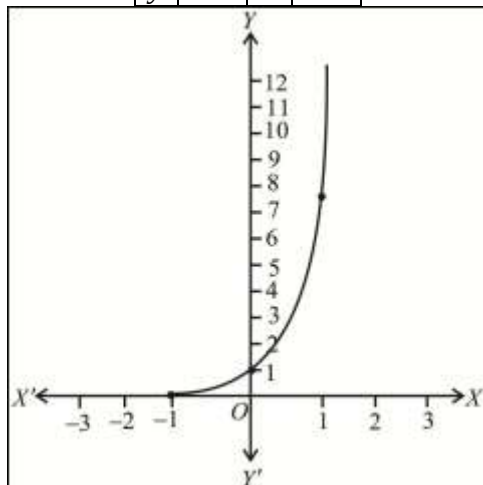


(iii) $y = e^{2x}$

Solution:

Given $y = e^{2x}$

x	-1	0	1
y	0.1	1	7.4

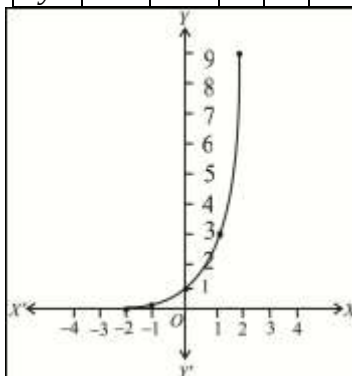


(iv) $y = 3^x$

Solution:

$y = 3^x$

x	-2	-1	0	1	2
y	0.1	0.3	1	3	9



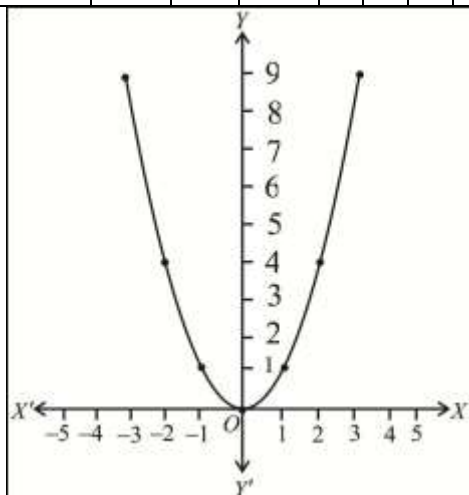
Q.2 Graph the curves that has the parametric equations given below

(i) $x=t, y=t^2, -3 \leq t \leq 3$ where “ t ” is a parameter

Solution:

$$x=t, y=t^2, -3 \leq t \leq 3$$

t	-3	-2	-1	0	1	2	3
$x=t$	-3	-2	-1	0	1	2	3
$y=t^2$	9	4	1	0	1	4	9

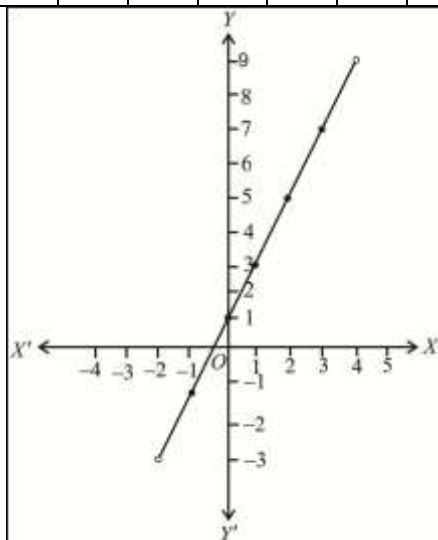


(ii) $x=t-1, y=2t-1, -1 < t < 5$ where “ t ” is a parameter

Solution:

$$x=t-1, y=2t-1, -1 < t < 5$$

t	-1	0	1	2	3	4	5
$x=t-1$	-2	-1	0	1	2	3	4
$y=2t-1$	-3	-1	1	3	5	7	9



(ii) $x=\sec \theta, y=\tan \theta$ where “ θ ” is a parameter

Solution:

$$x=\sec \theta, y=\tan \theta$$

$$\Rightarrow x^2 - y^2 = \sec^2 \theta - \tan^2 \theta$$

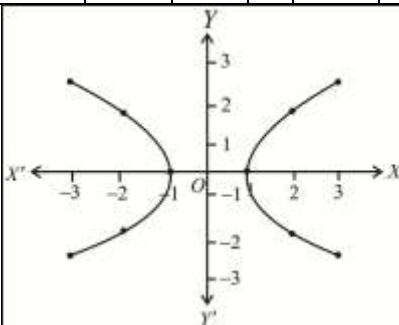
$$x^2 - y^2 = 1$$

$$x^2 - 1 = y^2$$

$$y = \pm\sqrt{x^2 - 1}$$

Here domain = $(-\infty, -1] \cup [1, \infty)$

x	-3	-2	-1	1	2	3
y	± 2.8	± 1.7	0	0	± 1.7	± 2.8



Q.3 Draw the graphs of the functions defined below and find whether they are continuous.

(i) $y = \begin{cases} x-1 & \text{if } x < 3 \\ 2x+1 & \text{if } x \geq 3 \end{cases}$

Solution:

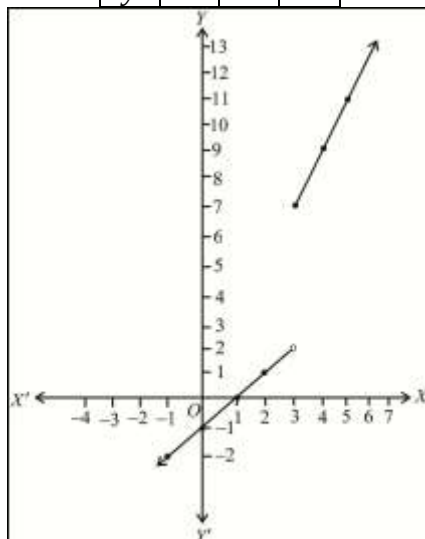
$$y = \begin{cases} x-1 & \text{if } x < 3 \\ 2x+1 & \text{if } x \geq 3 \end{cases}$$

Table for $y = x-1, x < 3$

x	-1	0	1	2	3
y	-2	-1	0	1	2

Table for $y = 2x+1, x \geq 3$

x	3	4	5
y	7	9	11



(ii) $y = \frac{x^2 - 4}{x - 2}, x \neq 2$

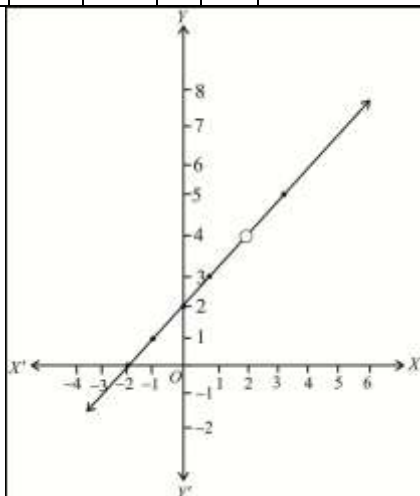
Solution:

$$y = \frac{x^2 - 4}{x - 2}, x \neq 2$$

$$y = \frac{(x+2)(x-2)}{(x-2)}$$

$$y = x + 2, x \neq 2$$

x	-2	-1	0	1	2	3
y	0	1	2	3	Undefined	5

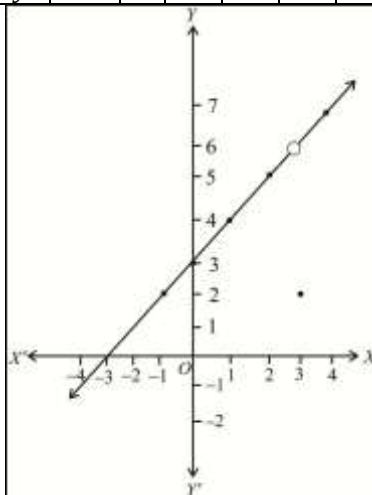


(iii) $y = \begin{cases} x+3 & \text{if } x \neq 3 \\ 2 & \text{if } x = 3 \end{cases}$

Solution:

$$y = \begin{cases} x+3 & \text{if } x \neq 3 \\ 2 & \text{if } x = 3 \end{cases}$$

x	-1	0	1	2	3	4
y	2	3	4	5	2	7



(iv) $y = \frac{x^2 - 16}{x - 4}, x \neq 4$

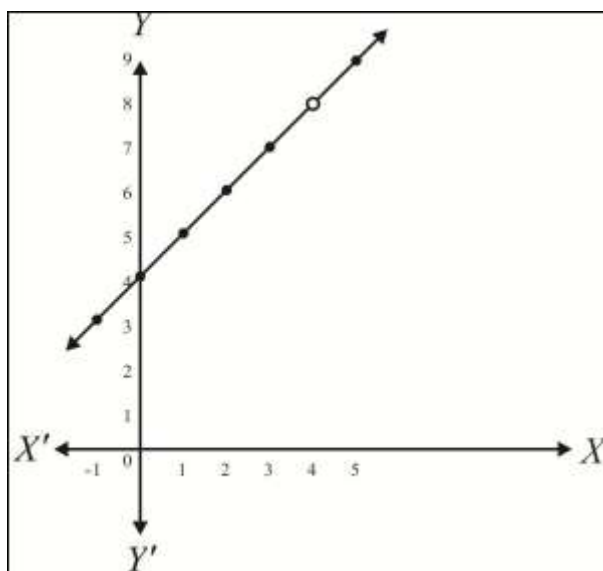
Solution:

$$y = \frac{x^2 - 16}{x - 4}, x \neq 4$$

$$y = \frac{(x+4)(x-4)}{(x-4)}, x \neq 4$$

$$y = x + 4, x \neq 4$$

x	-1	0	1	2	3	4	5
y	3	4	5	6	7	Undefined	9



Q.4 Find the graphical solution of the following equations:

(i) $x = \sin 2x$

Solution:

Let $y = x = \sin 2x$

So $y = x$

x	0°	90°
y = x(radian)	0	1.6

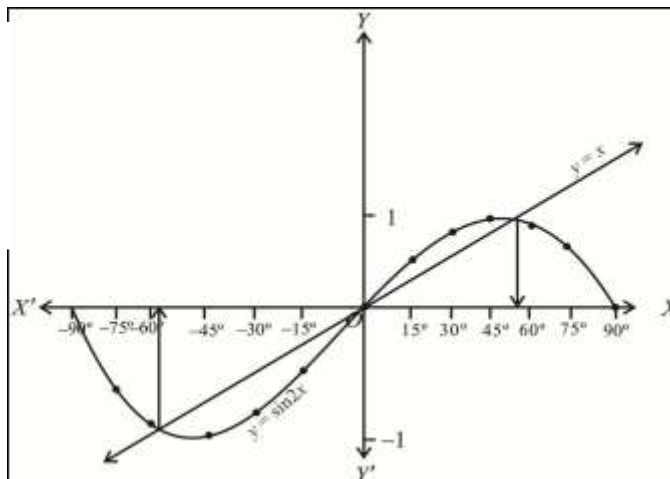
Also $y = \sin 2x$

X	-90°	-75°	-60°	-45°	-30°	-15°	0°	15°	30°	45°	60°	75°	90°
y = sin 2x	0	-0.5	-0.6	-1	-0.9	-0.5	0	0.5	0.9	1	0.9	0.5	0

From two graphs, solutions are

$$x = -55^\circ, 0^\circ, 55^\circ$$

$$\text{Solution set} = \{-55^\circ, 0^\circ, 55^\circ\}$$



(ii) $\frac{x}{2} = \cos x$

Solution:

Let $y = \frac{x}{2} = \cos x$

So $y = \frac{x}{2}$

x	0°	60°
$y = \frac{x}{2}$ (radian)	0	0.5

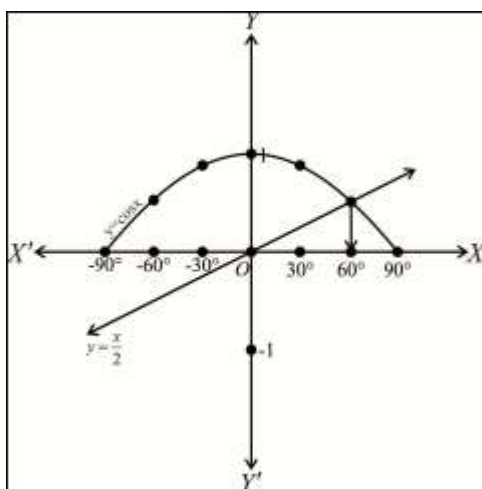
Also $y = \cos x$

x	-90°	-60°	-30°	0°	30°	60°	90°
$y = \cos x$	0	0.5	0.9	1	0.9	0.5	0

From two graphs, solution is

$$x = 60^\circ$$

$$\text{Solution set} = \{60^\circ\}$$



(iii) $2x = \tan x$

Solution:

Lets $y = 2x = \tan x$

So $y = 2x$

x	0°	60°
$y = 2x$ (radian)	0	2.1

Also $y = \tan x$

x	-90°	-60°	-30°	0°	30°	60°	90°
$y = \tan x$	Undefined	-1.7	-0.6	0	0.6	1.7	Undefined

From two graphs, solution is
 $x = 0^\circ$

Solution set = $\{0^\circ\}$

